Shadow-Loom: Causal Reasoning over Graphical World Models of Narratives

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Abstract

Stories hold a reader's attention because they have causes, secrets, and consequences. **Shadow-Loom** is an experimental open-source framework that turns a narrative into a versioned *graphical world model* and lets two engines act on it: a *causal physics* grounded in Pearl's ladder of causation (Pearl, 2009; Bareinboim et al., 2022) and the recently proposed counterfactual calculus over Ancestral Multi-World Networks (Correa and Bareinboim, 2025); and a *narrative physics* that scores the same graph against four structural reader-states — mystery, dramatic irony, suspense, and surprise — in the tradition of Sternberg's curiosity / suspense / surprise triad (Sternberg, 1992), with suspense formalised in the structural-affect line of Brewer and Lichtenstein (1982), Comisky and Bryant (1982) and Cheong and Young (2015). Large language models are used only at the boundary: extraction, rendering, and audit; identification, intervention, and counterfactual reasoning are carried out in typed code over the graph. The system is offered as a research artefact rather than as a benchmarked NLP model; code, fixtures, and pipeline are released open source.

1 Introduction

Most current LLM-driven story systems treat a narrative as a sequence-prediction problem and absorb the causal, temporal, and epistemic structure of the story into the weights of a single model. Recent work shows that such models are competent at *reading* causal relations from text (Kıcıman et al., 2024) but markedly weaker at *acting* on the implicit beliefs of characters they have just inferred (Gu et al., 2026; Kim et al., 2023), and similarly weak when asked to reason hypothetically about other agents' strategies without external scaffolding (Cross et al., 2024); in unconditioned generation they tend to collapse onto a small set of shared plot tropes (Xu et al., 2025; Tian et al., 2024). The complementary direction explored here keeps an explicit, typed graph of the storyworld (Schank and Abelson, 1977; Mani, 2012; Elson, 2012) and performs identification, intervention, and counterfactual reasoning over it symbolically (Pearl, 2009; Halpern, 2016; Correa and Bareinboim, 2025), using the LLM only as a constrained renderer (Riedl and Young, 2010; Yao et al., 2019; Goldfarb-Tarrant et al., 2020; Yang et al., 2023). Concretely, a query such as "what if Macbeth refuses to murder Duncan?" becomes a rung-2 (Intervention) do-operation on a typed event graph extracted from the play (App. B.2) rather than a free-form prompt to a language model; "does Poirot know the conspirators are still lovers?" becomes a finite belief-set membership test (App. B.5) rather than a single forward pass.

Motivation 1: counterfactuals are where LLMs tend to struggle. Pearl's hierarchy is provably non-collapsible. Rung-3 quantities (Counterfactuals) are not, in general, identifiable from rung-2 (Intervention) data, let alone from rung-1 (Observation) data (Bareinboim et al., 2022). Empirically, LLMs reproduce surface causal patterns from their training corpus (Kıcıman et al., 2024) but tend to degrade sharply on multi-hop belief-conditioned action prediction (Kim et al., 2023; Gu et al., 2026) and on story-internal counterfactual consistency (Xu et al., 2025). The questions that ordinarily matter to a reader of a story — "what if Macbeth had refused?", "what if Roy had told Mary the truth?" — live on the rung where the modern foundations (AMWN + ctf-calculus, Correa and Bareinboim, 2025) are designed to operate, and where end-to-end neural generators are not. A typed graph plus an explicit do-operator and abduction step turns each such question into a finite sequence of well-defined operations (Pearl, 2009; Halpern and Pearl, 2005; Halpern, 2016) rather than a free-form imagination task.

Motivation 2: coding agents do better, arguably, because they compile and test. A familiar pattern in agent design is that coding agents tend to outperform open-domain agents at fixed model capacity (Jimenez et al., 2024). Their outputs can be compiled, executed against a test suite, and rejected when they fail. The lever, on this reading, is the *typed substrate underneath the natural-language reasoning*, not the model itself. Shadow-Loom borrows that lever for narrative. A continuation is rejected if any of three checks fails: (a) its causal propagation (Eq. 2) violates inertia or affordance gating; (b) its counterfactual branch fails an AMWN-style consistency, independence, or exclusion check (Correa and Bareinboim, 2025); or (c) its re-extraction does not match the brief that licensed it (the audit pass, in the LLM-as-judge tradition of Bai et al., 2022; Madaan et al., 2023; Gu et al., 2024). The graph plays something like the role of the compiler; the auditor plays something like the role of the test suite.

The framework combines four ideas which, so far as I can tell, have not previously been wired together in a single open-source pipeline:

1. A **Pydantic-typed world graph** (`WorldStateV1`) with entities, events, beliefs, locations, channels, and three kinds of edge (causal, social, spatial), every node and edge tagged with both a chronological *fabula* time and a narrative *syuzhet* index (Shklovsky, 1965; Genette, 1980).

2. A **causal physics engine** — a graphical world model of cause and effect — that implements Pearl's three rungs — observation, do-intervention, and abductive counterfactual — with a propagation rule gated by an explicit "Impact > Inertia" threshold and spatial affordance checks.

3. A **narrative physics layer** — a graphical world model of reader affect over the same graph — that scores it against closed-form information-state functionals for mystery, dramatic irony, suspense, and surprise. Suspense uses the hope/fear (here *hope/threat*) anticipation pair from the structural-affect tradition (Brewer and Lichtenstein, 1982; Comisky and Bryant, 1982; Ortony et al., 1988; Zillmann, 1996) as operationalised by Cheong and Young (2015), combined as a *balance* $\times$ *stakes* product over the unrevealed ledger; this is a deliberate departure from the information-theoretic uncertainty-reduction formulation of Wilmot and Keller (2020), whose reader-vs-future-state epistemic-asymmetry shape we instead borrow for the dramatic irony scorer in §A.5. Surprise is binary KL divergence between a leave-one-out corpus-marginal prior, evidence-updated by revealed causal edges, and the entity's reconstructed final state.

4. A **constrained-LLM renderer + recursive auditor** loop (Gu et al., 2024) that converts the simulation result into a typed creative brief, generates prose under that brief, and then re-extracts the prose to verify no "miracle steps" were introduced.

The system is named for the dual-graph metaphor: a *loom* of factual edges runs in parallel with one or more *shadow* counterfactual branches, in the sense of the AMWN construction of Correa and Bareinboim (2025). Section 2 sketches the pipeline; Section 4 states what is novel and what is borrowed; Section 5 gives the design rationale; Section 6 discusses relevance to NLP, reasoning, and social science; Section 7 states the limitations honestly. Equations and per-stage definitions are in App. A; an extended narrative walkthrough of the whole loop on a single fixture is in App. C; worked examples drawn from the twenty bundled plot fixtures follow in App. B; and a separate appendix on the authoring user interface, with screenshots and step-by-step example sessions, is in App. D.

How to read this paper. The main text is intentionally short. It states the design, situates it in the literature, and records what is novel as a combination. Readers who want to understand the symbolic machinery in detail — the schema, the ego-graph extraction, the do-operator, the abductive counterfactual, the four affective scorers, the brief-and-audit loop — should treat App. A as the authoritative specification. App. C is a long-form description of the same pipeline expressed as a story: *Macbeth* is fed in as raw prose and followed step-by-step through ingestion, intervention, propagation, scoring, rendering, and audit, with every intermediate object shown. The eight example sub-appendices in App. B then show how each individual stage manifests on a different bundled fixture. App. D is for readers who want to understand or use the authoring workspace; it cross-links every UI surface to the pipeline stage it exposes and includes example sessions on real fixtures.

2 Pipeline overview

The pipeline is a five-phase loop over a versioned world model. Every version is ancestor-linked, so a counterfactual fork is literally a sibling row in a directed acyclic version tree, not a mutation of canon. Figure 1 sketches the loop.

`prose`
$\downarrow$ ingestion (5-pass extraction)
`WorldStateV1 (typed graph)`
$\downarrow$ ego-graph slice + AMWN sandbox
`NetworkX MultiDiGraph`
$\downarrow$ causal physics (do, abduction, propagation)
`candidate sandbox states`
$\downarrow$ narrative physics (affective scoring)
`CreativeBrief (typed constraints)`
$\downarrow$ constrained LLM render
`prose`
$\downarrow$ recursive audit + re-extraction
`new VersionRow (factual or shadow)`

Figure 1: The Shadow-Loom loop. The LLM is invoked only at the top (ingestion), the bottom (rendering), and the audit; everything in between is symbolic computation over the graph. See the appendix for definitions of each stage.

The five phases correspond to the five appendix subsections (A.2–A.7): **(1) Ingestion** runs five LLM agents to extract a typed `GlobalRegister`, then three concurrent specialist agents (physics, social, consequences) per chunk to extract events, relationships, channels, and entity-state deltas, with a programmatic validation and auto-repair loop. **(2) Causal physics** forks an Ancestral Multi-World Network sandbox, applies a do-operator for interventions, runs abduction for counterfactuals, and propagates effects through a topological sort of the causal sub-graph subject to inertia and affordance gating. **(3) Narrative physics / affective calculus** scores each candidate sandbox against four structural information-state functionals and six trait-trajectory emotional functionals. **(4) Generation** packages the highest-scoring candidate as a `CreativeBrief` of typed `ConstraintBlocks` and hands *only* the brief to a creative LLM. **(5) Audit and feedback** runs an LLM-as-judge over the prose, detects "miracle steps" (state changes with no licensing edge in the brief), and either accepts the version or re-runs generation with the auditor's feedback appended, up to a configurable retry budget.

The entire loop is exposed both through a NiceGUI authoring workspace (nine cross-linked tabs over a versioned project DAG; App. D) and through a Model-Context-Protocol server with roughly forty tools, so a downstream agent can drive shadow-branch creation, scoring, and promotion programmatically.

3 Theoretical lineage and computational counterparts

The system has been built theory-first: every named field in `WorldStateV1` corresponds to a specific narratological, cognitive, or causal-inference construct, and every field has a prior computational instantiation that the design either generalises or borrows from. The pairings the design depends on are summarised below; an extended mapping lives in the project's academic-foundations document.

Fabula vs syuzhet (Shklovsky, 1965; Tomashevsky, 1965; Genette, 1980; Bordwell, 1985; Bal, 2009). Two integer indices on every `EventNode`: chronological $f(e)$ and presentation $s(e)$. The four affective scorers reduce to set operations between the two projections (App. A.5).

Possible-worlds narratology (Ryan, 1991, 2001; Doležel, 1998; Pavel, 1986), modal-logical roots (Lewis, 1973), and counterfactual graphical models (Balke and Pearl, 1994; Shpitser and Pearl, 2008; Correa et al., 2021; Correa and Bareinboim, 2025). Every node and edge carries `world_id` $\in \{\text{factual}, \text{shadow}\}$; counterfactual queries fork a sibling `VersionRow`. The AMWN tag and three-rule mental model (consistency, independence, exclusion) come from Correa and Bareinboim (2025); I adopt the naming and architecture even though the typed narrative graph used here is not a rigorous SCM. AMWN supersedes the Twin-Network construction (Balke and Pearl, 1994) and the multi-network machinery of Shpitser and Pearl (2008), and is the counterfactual-completeness counterpart to do-calculus (Pearl, 1995, 2009; Bareinboim et al., 2022).

Greimas (Greimas, 1983), Propp (Propp, 1968), Bremond (Bremond, 1973), and the symbolic story-grammar tradition (Rumelhart, 1975; Mandler and Johnson, 1977; Kintsch and van Dijk, 1978; Schank and Abelson, 1977; Lehnert, 1981; Meehan, 1977; y Pérez and Sharples, 2001; Bringsjord and Ferrucci, 2000; Chambers and Jurafsky, 2008). `GlobalTrait` (`WORLD_`) realises Greimas's "Power" actant;

`RelationshipEdge.power_dynamic` mirrors the sender / receiver / helper / opponent topology; `EventNode.event_type` $\in$ {choice, outcome, revelation, utterance} is a coarse generalisation of Propp's functions, closer in spirit to Bremond's triad. The plot-unit and script traditions prefigure both the typed-event graph (Schank and Abelson, 1977; Lehnert, 1981) and the goal-directed planner / generator pipeline (Meehan, 1977; y Pérez and Sharples, 2001); the unsupervised narrative event chains of Chambers and Jurafsky (2008) are a direct precursor to the typed `causal_topology` maintained at runtime.

Cognitive narratology (Herman, 2002; Fludernik, 1996; Zwaan and Radvansky, 1998; Gerrig, 1993) and constructionist comprehension (Trabasso and van den Broek, 1985; Graesser et al., 1994; Kintsch and van Dijk, 1978). The five fields of the situation-model (time, space, protagonist, causation, intention) are nearly the field set on `EventNode`; that readers materialise causal links explicitly during comprehension is the empirical justification for materialising them explicitly in storage.

Theory of mind (Premack and Woodruff, 1978; Wimmer and Perner, 1983; Baron-Cohen et al., 1985; Zunshine, 2006; Baker et al., 2009; Chandra et al., 2024), its modern LLM benchmarks (Kim et al., 2023; Gu et al., 2026), and ToM-scaffolded LLM agents (Cross et al., 2024). `Belief{value, confidence, inertia, established_at_fabula, acquired_via_event_id, acquired_via_channel_id}` is a discrete approximation of recursive ToM with provenance. The benchmarks above motivate the auditor's separate behaviour-given-belief consistency check.

Sternberg's curiosity / suspense / surprise triad (Sternberg, 1978, 1992; Brewer and Lichtenstein, 1982; Baroni, 2007) and its computational realisations (Cheong and Young, 2015; Bae and Young, 2008; Ely et al., 2015; Wilmot and Keller, 2020, 2021; Wilmot, 2022). The four structural scorers (MYS, IRO, SUSP, SUR) extend the triad with dramatic irony (Booth, 1974; Gerrig, 1993). The suspense scorer follows the hope/fear anticipation pair of the structural-affect and appraisal traditions (Brewer and Lichtenstein, 1982; Comisky and Bryant, 1982; Ortony et al., 1988; Zillmann, 1996; Gerrig, 1989) as operationalised in narrative planning by Cheong and Young (2015), combining the two sides as a *balance* $\times$ *stakes* product that peaks under genuine outcome uncertainty rather than under one-sided dominance. This is a deliberate departure from Wilmot and Keller (2020), whose suspense is information-theoretic uncertainty reduction over neural state representations and whose decision-theoretic counterpart is Ely et al. (2015); that reader-vs-future-state epistemic-asymmetry shape (the reader knows things the focal future does not yet contain) is structurally closer to dramatic irony (reader knows things the characters do not) than to a hope/threat ledger, and motivates the irony scorer in App. A.5 rather than the suspense scorer. Surprise follows Itti and Baldi (2009) with a geometric prior update over a leave-one-out corpus marginal; emotional arc analytics owe to Reagan et al. (2016).

Information-theoretic and epistemic substrate (Shannon, 1948; Fagin et al., 1995; Alchourrón et al., 1985). `Channel` carries `intelligibility` $\in [0, 1]$ per participant (generalising the legacy encryption boolean) and an `eavesdropped_by` relation derived from the `intelligibility_threshold`; belief revision uses an AGM-style `beliefs_added / beliefs_invalidated` delta.

Causal inference (Pearl, 1995, 2009; Pearl and Mackenzie, 2018; Spirtes et al., 2000; Bareinboim et al., 2022) and actual causality (Halpern and Pearl, 2005; Halpern, 2016; Woodward, 2003). Pearl's three rungs — **rung 1 Correlation** (observation), **rung 2 Interventions** (do($\cdot$)), and **rung 3 Counterfactuals** (abduction + do) — map directly to the query taxonomy used here (`ObservationQuery`, `InterventionQuery`, `CounterfactualQuery`); `mechanism` and `causal_force` on edges are motivated by Halpern's "actual cause"; ego-graph slicing is a heuristic Markov-blanket cut (Verma and Pearl, 1988). Recent benchmarking of LLMs as causal reasoners (Kıcıman et al., 2024) supports the hybrid stance taken here: LLMs propose edges (`extract_graph.py`); typed code identifies and propagates them (`causal_physics.py`).

Plan-and-render era of neural story generation (Riedl and Young, 2010; Jhala and Young, 2010; Martin et al., 2018; Yao et al., 2019;

Goldfarb-Tarrant et al., 2020; Rashkin et al., 2020; Yang et al., 2022, 2023) and its benchmarks (Mostafazadeh et al., 2016; Fan et al., 2018; Akoury et al., 2020; Tian et al., 2024; Xu et al., 2025). The `CreativeBrief` (Step 9) plus channel-aware draft (Step 10) is a typed, causal-graph-conditioned variant of the same plan-then-render pattern. The brief enumerates per-effect typed `ConstraintBlocks`; the auditor (Step 11) is the rescore / revise step (Goldfarb-Tarrant et al., 2020; Yang et al., 2022) specialised to narrative-causal rather than stylistic criteria, in the LLM-as-judge tradition (Zheng et al., 2023; Bai et al., 2022; Madaan et al., 2023; Gu et al., 2024; Qi et al., 2023), with structured prompting and self-improvement techniques (Wei et al., 2022; Zelikman et al., 2022) used inside the renderer when free-form reasoning is required.

Reading as simulation (Mar and Oatley, 2008; Oatley, 2016, 1995; Hogan, 2003). The reason a tight causal/affective model matters more than surface fluency: fiction is a cognitive simulator, and what is being simulated is a typed world.

Game engines, interactive narrative, and world models for AI (Mateas and Stern, 2003; Riedl and Bulitko, 2013; Ware and Young, 2014; Kreminski et al., 2020; Park et al., 2023; Ha and Schmidhuber, 2018; LeCun, 2022; Hao et al., 2023; Wang et al., 2024; Smelik et al., 2014; Summerville et al., 2018). Architecturally, Shadow-Loom is closer to a game engine than to a sequence model: a typed simulation step (causal physics) is wrapped by a director / drama-manager (narrative physics) and exposed to a stochastic policy (the LLM renderer) only through a constrained action interface, in the lineage of Mimesis / Façade-style interactive drama (Mateas and Stern, 2003; Riedl and Bulitko, 2013) and narrative planners such as CPOCL/Glaive (Ware and Young, 2014). The generative-agent line of work (Park et al., 2023) and the "world model" line in deep RL (Ha and Schmidhuber, 2018; LeCun, 2022; Hao et al., 2023) make a similar wager from the opposite direction: that an explicit, manipulable latent world is what makes long-horizon coherent behaviour tractable. Shadow-Loom's shadow branches play the role those world models play — a sandbox in which counterfactual rollouts can be evaluated before being committed — but with a typed, inspectable schema in place of an opaque latent state. The bundled example worlds and authoring tools relate Shadow-Loom to the broader procedural-content-generation tradition (Smelik et al., 2014; Summerville et al., 2018; Kreminski et al., 2020).

4 What is novel, what is borrowed

Borrowed. Every ingredient in Section 3 is borrowed: Pearl's ladder (Pearl, 2009; Bareinboim et al., 2022); the AMWN + ctf-calculus framework (Correa and Bareinboim, 2025); actual-causality semantics (Halpern, 2016); the structural-affect / hope-fear anticipation pair underlying the suspense scorer (Brewer and Lichtenstein, 1982; Comisky and Bryant, 1982; Ortony et al., 1988; Zillmann, 1996; Cheong and Young, 2015); the uncertainty-reduction view of suspense (re-used here as the epistemic-asymmetry shape of the irony scorer) of Wilmot and Keller (2020); the Bayesian-surprise formalism of Itti and Baldi (2009); the LLM-as-judge audit pattern (Bai et al., 2022; Madaan et al., 2023; Gu et al., 2024); the plan-and-render line in neural story generation (Riedl and Young, 2010; Yao et al., 2019; Goldfarb-Tarrant et al., 2020; Yang et al., 2023, 2022; Rashkin et al., 2020); the typed-event / story-grammar tradition (Schank and Abelson, 1977; Lehnert, 1981; Mani, 2012; Elson, 2012); and the narratological foundations (Shklovsky, 1965; Genette, 1980; Sternberg, 1992; Ryan, 1991; Herman, 2002; Zwaan and Radvansky, 1998).

Novel as a combination. I am not aware of another open-source pipeline that, taken together: (a) runs all three Pearl rungs over a typed narrative graph; (b) tags every node and edge with an AMWN `world_id` (Correa and Bareinboim, 2025) so that counterfactuals are first-class persisted objects; (c) couples that machinery to closed-form scorers for the four classical reader-states of narrative cognition; (d) uses the resulting score as a loss function over candidate interventions, with the LLM constrained to render the winner; and (e) closes the loop by re-extracting the rendered prose and comparing it against the brief. Each ingredient is borrowed. The composition is the modest contribution: a typed two-engine substrate underneath a constrained renderer. Architecturally the closest analogue is something like a game engine for narrative: a typed simulation step gated by a drama-manager-like scorer (Mateas and Stern, 2003; Riedl and Bulitko, 2013; Ware and Young, 2014), driving a constrained stochastic actor in the spirit of recent generative-agent and world-model work (Park et al., 2023; Ha and Schmidhuber, 2018; Hao et al., 2023), but with the latent state replaced by an explicit causal-graphical schema.

Specific small constructions. The "Impact > Inertia" propagation rule (Eq. 2) and the geometric prior update inside the surprise scorer (Eq. 16) are introduced here as deliberately simple choices that keep both physics monotonic in useful ways (App. A.5). The two-axis fabula/syuzhet sampling of the affective scorers (App. A.5) is, to my knowledge, also new.

5 Rationale for the design choices

Why a typed graph rather than free text + RAG? Recent stress tests show that LLMs can answer "what does X believe?" but fail to behave consistently with that belief in downstream action prediction (Gu et al., 2026; Kim et al., 2023). Long-context reading itself is unreliable in the same regime (Liu et al., 2024), which compounds the problem when the "world" has to be reconstructed from a growing prose buffer. Belief and causal structure that are recoverable from prose are not, on present evidence, reliably *operated on* inside the prose-only loop. A typed graph makes belief, channel intelligibility, and causal mechanism inspectable and intervenable.

Why both a causal and a narrative physics? The causal physics decides *what is allowed to happen* given the graph; the narrative physics decides *which of the allowed things would be most readable*. Decoupling licensing from desirability lets the system score and rank candidate interventions before any text is generated, and lets the same engine serve both "what-if" analytical queries (rung-2 Intervention and rung-3 Counterfactual) and high-level affective directives ("maximise dramatic irony for character X").

Why constrain the LLM rather than fine-tune one? The renderer's job is reduced to dialogue, description, and pacing within a mathematical envelope. Whatever the model's training distribution, the brief enumerates the causal edges, beliefs to preserve, channels to keep hidden, and trait shifts to honour. Errors that survive into prose are then catchable by the audit pass because they correspond to specific, testable claims — the "compile and test" analogue from Motivation 2.

Why AMWN tagging at the persistence layer? Treating a counterfactual as a sibling `VersionRow` with `world_id="shadow"` makes "what-if" branches first-class, diffable, promotable, and auditable; it is the persistence-layer analogue of the AMWN graphical construction (Correa and Bareinboim, 2025). A user can compare a shadow against canon, keep both, or promote.

6 Why this might matter beyond fiction

NLP / reasoning. Narrative is, for these purposes, a reasonably tractable test bed for counterfactual and theory-of-mind reasoning, because the ground truth (what happens, who knows what) is finite and authored. A typed-graph + symbolic physics + constrained-renderer architecture is one concrete instantiation of the neuro-symbolic stance for controllable generation; the audit pass produces a falsifiable record of where the LLM departed from the brief. Recent ToM benchmarks (Gu et al., 2026; Kim et al., 2023) and ToM-scaffolded LLM agents (Cross et al., 2024) ask exactly the kind of behaviour-given-belief questions that an explicit `Belief` field with provenance is designed to support.

Computational social science and digital humanities. Many socio-historical and humanistic questions are counterfactual in form ("what if policy X had not been adopted?", "how would this novel read if the protagonist had spoken sooner?"). A graphical substrate that lets a researcher ingest a narrative source, fork a shadow branch under an explicit intervention, and inspect the propagated trait and relationship deltas is a small step toward auditable narrative simulation that can sit alongside established computational-social-science (Lazer et al., 2009) and digital-humanities tooling. I make no claim of empirical validity for any specific simulation; the contribution is the substrate, not its calibration.

Author tooling. The same machinery surfaces in a NiceGUI workspace and a Model-Context-Protocol server with around forty tools, exposing the graph, the affective scorers, and the version DAG to either human authors or LLM agents. The workspace is described component-by-component in App. D.

An invitation to other research communities. The substrate is deliberately domain-agnostic at the graph level: entities, events, beliefs, channels, and counterfactual branches are not specific to fiction. Computational social scientists (Lazer et al., 2009) routinely treat narrative sources — court records, parliamentary debates, oral histories, news streams — as data, and digital-humanities scholarship has long sought structured, sharable representations of literary works that can support both close and distant reading. Shadow-Loom may be read as one such representation: an open, typed, version-controlled graph with an explicit counterfactual sandbox and an audit trail back to the source text. Researchers in those communities, as well as in narratology and cognitive science, are warmly invited to fork the schema, swap in their own ingestion prompts and affective scorers, and report what they find. The pipeline ships with twenty worked example worlds drawn from canonical literature and film, which may serve as a common reference set for cross-group experimentation.

7 Limitations and future evaluation

Shadow-Loom is an experimental research artefact rather than a benchmarked NLP system, and I want to be explicit about what that does and does not imply. No headline numbers are reported against an external dataset: the scores produced by the narrative physics are design choices grounded in narratological theory rather than validated reader-reaction models, and the causal physics borrows the AMWN naming and three-rule mental model of Correa and Bareinboim (2025) without implementing their identification algorithm literally — the graph is heavily typed for narrative use rather than for rigorous structural causal model identification. The ingestion pipeline depends on LLM extraction, and although a programmatic validation and auto-repair loop catches the more obvious failure modes, extraction errors can still propagate. The current unit-test suite covers the symbolic layers but does not amount to external evaluation.

Formal evaluation is therefore deferred to future work, along several complementary axes: (i) human-judgement studies of the affective scorers against reader annotations, in the spirit of Wilmot and Keller (2020); (ii) targeted ablations on bundled fixtures to quantify the contribution of the causal physics, the AMWN sandbox, and the audit loop to downstream coherence; and (iii) comparison against end-to-end LLM baselines on counterfactual and theory-of-mind benchmarks (Kim et al., 2023; Gu et al., 2026; Cross et al., 2024; Xu et al., 2025). I share the present design as a substrate for further research — by myself and by others — not as a finished product.

As an internal sanity check, the four structural-affect scorers were audited against the bundled twenty-fixture corpus (`example_worlds/`) at seven evenly-spaced syuzhet anchors per fixture, giving 162 score evaluations per metric (one fixture emits a flat-zero suspense column at its terminal anchor and so contributes only six points). The per-scorer scale summary is mystery $0.17/0.53/1.00$ (min/median/max), dramatic-irony $0.00/0.45/0.90$, suspense $0.00/0.16/0.39$, surprise (per-step local mode) $0.00/0.03/0.24$ — the four scorers occupy different absolute bands by design (mystery is a population fraction near 1 early in the syuzhet; surprise is a per-step KL spike near 0 between revelation points). The audit confirms the canonical signatures the underlying theory predicts: monotone mystery decline across every world; rise-peak-fall irony arcs in $16/20$ worlds (Macduff hearing of his family, Poirot's denouement, Nick's letter to Daisy); terminal-anchor suspense discharge in every world (no unrevealed threats remain); surprise spikes only at canonical revelation points (Reservoir Dogs Mr Orange flashback 0.19, Gone Girl mid-novel perspective shift 0.20, Wuthering Heights in-medias-res frame 0.24). All scorer constants are externalised through `DirectiveAssemblySettings` (env-var prefix `DIRECTIVE_ASSEMBLY_*`); the audit script `scripts/audit_affective.py` reproduces the table.

Code, licence, and reproducibility

The code is released open source under AGPL-3.0-or-later, with a parallel commercial licence; the authoritative LICENSE file lives in the project repository. The pipeline, schema, MCP surface, auditor, and the affective scorers described here all sit in the `shadow_loom/` package. Worked examples on bundled plots (Macbeth, Death on the Nile, Reservoir Dogs, and others) ship in `example_worlds/` and `sample_plots/`.

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A Definitions and equations

This appendix gives the formal definitions of the data model and the equations of each pipeline stage. Section references in the main paper point here.

A.1 World model schema (**`WorldStateV1`**)

A world state is a tuple

$$\mathcal{W} = (\mathcal{N}, \mathcal{E}, \mathcal{C}, \mathcal{T}, \tau)$$

with nodes $\mathcal{N}$ (entities ENT_•, events EVT_•, locations LOC_•, objects OBJ_•, channels CHN_•, world traits WORLD_•), edges $\mathcal{E}$ (causal, relationship, spatial), channels $\mathcal{C}$ with per-participant intelligibility $\iota \in [0, 1]$, world traits $\mathcal{T}$, and a branch tag $\tau \in \{\text{factual}, \text{shadow}\}$ inherited by every node and edge.

Each event $e \in \mathcal{N}_{\text{EVT}}$ carries two integer indices: chronological $f(e) \in \mathbb{Z}$ (*fabula time*) and presentation $s(e) \in \mathbb{Z}$ (*syuzhet index*), with s contiguous and unique. Trait values, beliefs, and ambient state are carried by typed vectors $(v, \iota_{\text{inertia}}, \sigma_{\text{evidence}})$ representing value, inertia, and evidence strength. Beliefs carry provenance (*event_id*, *channel_id*) and the time they became established. A pure function $\textsc{Reconstruct}(e_i, t)$ replays sparse `state_timeline` deltas to give the entity's effective state at any fabula time t.

Node types and their carrying state. Table 1 summarises the six node families and the typed state each carries. All six share a common `AMWNNode` base that carries the branch tag τ (`world_id`), a syuzhet-indexed `state_timeline`, and `evidence_strength`.

Node ID prefix	Pydantic class	Distinctive state
`ENT_`	`Entity`	`traits: Dict[str, TraitVector]`; `beliefs: List[Belief]`; `constants: List[str]`; `location_id`; `status`; `state_timeline` of `EntityStateSnapshot` deltas.
`EVT_`	`EventNode`	`event_type` ∈ {`choice`, `outcome`, `revelation`, `utterance`}; `actor_ids`, `target_ids`; `fabula_time`, `syuzhet_index`; for `utterance`: `speaker_id`, `addressee_ids`, `via_channel_id`, `content`, `truth_value` ∈ {`true`, `false`, `unknown`, `performative`} (the last covers prophecies / vows / orders, which alone may carry future-fabula `target_ids`; see §A.2.1).
`LOC_`	`Location`	`ambient_state: Dict[str, AmbientVector]` (e.g. *tension*, *bustle*, *supernatural*); `capacity`; outgoing `SpatialEdges`.
`OBJ_`	`Object`	`location_id` or `owner_id`; `properties`; `affordances: List[Affordance]` (typed action/target gates).
`CHN_`	`Channel`	`medium`, `directionality`, `participant_ids`; `intelligibility: Dict[entity_id, float]` per-participant; `established_at_fabula`, `terminated_at_fabula`; `discovered_at_syuzhet` (optional, marks reader-facing channel reveals).
`WORLD_`	`GlobalTrait`	Standing pressure (e.g. `WORLD_REGENCY_RANK` in Persuasion, `WORLD_SOVIET_PENETRATION` in Tinker Tailor): a single value v, `inertia`, `evidence_strength`, plus a `state_timeline` of `WorldTraitSnapshots`.

Table 1: Node families in `WorldStateV1`. Each `ID_` prefix maps to one Pydantic class in `shadow_loom/models.py` and carries its own typed state. The same six prefixes are used as node-ID conventions throughout the codebase, the bundled fixtures, and the renderer's brief.

Edge families and their roles. Edges live on three orthogonal layers, each a separate Pydantic class with its own validator:

`CausalEdge` (event⇄event, event→state, state→event, state→state). A single class with five `causality_type` modalities — `chain_reaction`, `mutation`, `mutation_social`, affordance_gate, ambient_propagation — and a model validator that enforces source/target type matches modality. Each edge carries mechanism (e.g. *psychological*, *physical*, *social*, *epistemic*, *betrayal*), evidence_strength $\in$ {weak, moderate, strong}, causal_force $\in$ $[0, 10]$, a fabula timestamp, an optional propagation_delay, and — for mutation / mutation_social — a trait_target and signed trait_delta. mutation_social edges additionally carry a rel_counterpart_id so that a single event can shock a relationship axis (e.g. Lady Macbeth's *power_dynamic* against Macbeth) rather than a free-standing trait.

RelationshipEdge (entity⇄entity). One edge per ordered pair, with a per-axis metrics: Dict[axis, AxisMetric] where axis $\in$ {*affinity*, *fear*, *power_dynamic*}. Each axis owns its own value, inertia, evidence_strength, and last_updated_fabula. Flat back-compat properties expose the canonical *affinity* value as .value.

SpatialEdge (location→location). Optional is_locked plus barrier_item_id; unlocked edges contribute to the spatial-affordance path-check that gates cross-location causal influence (Eq. 2).

Communication is *not* an edge — standing capability lives on the Channel *node*, and discrete messages are first-class EventNodes with event_type= "utterance" pointing at a channel via via_channel_id. This is what makes the lying-narrator pattern (Amy's diary in *Gone Girl*, Pip's misattributed benefactor in *Great Expectations*, the Witches' equivocations in *Macbeth*) a single uniform construct: an utterance event with truth_value="false" broadcast through a channel whose addressees have non-zero intelligibility.

Worked illustration — a slice of *Macbeth*. The Inverness night reduces to a small, fully typed subgraph:

- **Entities.** ENT_MACBETH with traits ambition ($v = 0.7$, $\iota = 0.5$), courage (0.85, 0.7), guilt (0.1, 0.25), paranoia (0.2, 0.3); a Belief(target_id="ENT_WITCHES", confidence=0.4, inertia=0.35).
- **Object.** OBJ_BLOODY_DAGGERS with Affordance(action="kill", target_type="Entity") — the precondition gate the engine checks before EVT_DUNCAN_MURDER fires.
- **Location.** LOC_INVERNESS_CASTLE with ambient *tension* = 0.6. LOC_HEATH carries ambient_state{supernatural: 0.9, concealment: 0.7}, the source of the heath-night ambient-propagation edge that biases Macbeth's uptake of the prophecy belief.
- **Channel.** CHN_HEATH_PROPHECY with participants=[ENT_MACBETH, ENT_BANQUO, ENT_WITCHES], used by the utterance event EVT_UTT_PROPHECY_HEATH (truth_value = *performative* — the witches' prophecy is a future-positing speech-act, not a fact-claim, and may therefore reference future events such as EVT_MACBETH_CROWNED in its target_ids).
- **Causal sub-graph.** EVT_LADY_MACBETH_PERSUADES → EVT_DUNCAN_MURDER (chain_reaction, causal_force = 8.0); OBJ_BLOODY_DAGGERS → EVT_DUNCAN_MURDER (affordance_gate, 9.0); EVT_LADY_MACBETH_PERSUADES → ENT_LADY_MACBETH (mutation_social, *power_dynamic* against ENT_MACBETH, $\Delta = +0.4$, 7.0); EVT_BANQUO_GHOST → ENT_MACBETH (mutation_social, *fear* against ENT_BANQUO, $\Delta = +0.5$, 6.0); LOC_HEATH → ENT_MACBETH (ambient_propagation, *epistemic*).

Every value above is read directly from `example_worlds/macbeth.py`; the same six-prefix / three-edge / typed-state pattern carries through every bundled fixture, from a Regency drawing-room (*Persuasion*) to a Cold-War mole hunt (*Tinker Tailor Soldier Spy*). The fixture-level diversity of *ambient_state* tags, *mechanism* labels, *trait* dictionaries, and *intelligibility* maps is intentional: the schema is fixed, the populated typology is fixture-specific.

A.2 Ingestion

Given prose P, ingestion produces a $\mathcal{W}$ via:

1. Five extraction agents over P build a `GlobalRegister` of WORLD_, ENT_, LOC_, OBJ_ nodes plus a fuzzy alias table. A chunked-parallel *proposition catalogue* pass (Phase A3) then mints PROP_ ids for every fact-claim the story will need to track — shared targets that downstream utterances and beliefs reference rather than duplicating free-text `perceived_states`. Per-chunk catalogues are union-deduped using both exact-string and token-Jaccard (≥ 0.75) semantic match so cosmetically-different phrasings of the same claim collapse onto a single canonical PROP_ id; concern seeds whose proposition collapsed are rewired onto the canonical id rather than dropped. A second global pass (Phase A3b) then sees the merged proposition list plus the entire source text in a single window and emits the authoritative CCN_ *concern seed* set — one `(entity, proposition, polarity)` record per standing fear or desire — which the chunked stage tends to under-emit because each chunk only sees a slice of the character's arc.

2. For each chunk, a Socratic Who/What/Where/When/Why/How scaffold dispatches three concurrent specialist agents: *physics* (events, causal/spatial edges), *social* (relationship edges, channels, utterance `EVT_UTT_` events), *consequences* (authoritative entity-state deltas). A per-chunk *affect sub-stage* runs after consequences to wire `asserts_proposition_id` / `denies_proposition_id` / `resolves_proposition_ids` onto the chunk's events.

3. Output is normalised, fabula times are remapped to a uniform $\Delta t = 100$ spacing, and a programmatic validator + LLM correction loop enforces ID closure and type constraints. The validator includes a temporal-coherence rule on utterances: see §A.2.1.

4. Three post-assembly passes complete the world model: (a) *concern extraction* per entity (gap-filler that runs only for entities the catalogue's Phase A3b seeds and the per-chunk affect agent's `new_concern_seeds` did not already populate, so a single LLM call per uncovered entity captures any standing fears or desires missed upstream); (b) *belief proposition clustering* (an LLM clusters free-text `perceived_state` strings sharing a target into the catalogue's `PROP_` ids); (c) *audience entity synthesis* — `_maybe_synthesise_audience_entity` constructs an `ENT_AUDIENCE` entity carrying graded belief snapshots over each `Proposition` on the syuzhet timeline. `ENT_AUDIENCE` is what the propositional affect scorers (§**??**) read when computing reader-side suspense, surprise, irony, and mystery.

5. A deterministic *Phase C′* post-pass layer (`apply_post_pass_fixes`) closes documented LLM gaps with no model cost: lexical EVT_$\rightarrow$PROP_ binding (token-overlap ≥ 3 plus ENT_/OBJ_ overlap) appends matching event ids onto every `event_occurs` / `outcome` proposition with empty `EVT_` referents; `truth_at_fabula` is then synthesised from the earliest referent event's fabula time; resolved concerns are auto-closed; and a per-concern *salience trajectory* pass walks events touching each concern's proposition and emits a salience spike at each touch plus a decay snapshot at the proposition's earliest truth commit, restoring per-concern dynamic range when the per-chunk affect agent skipped a chunk or emitted no concern snapshots.

A.2.1 Utterance temporal coherence

An `EventNode` with `event_type="utterance"` carries a `target_ids` list naming the entities, objects, and events the speech-act is *about*. The schema enforces a temporal-coherence invariant: every EVT_ id appearing in an utterance's `target_ids` must satisfy $\texttt{fabula\_time}_{\text{target}} \leq$

`fabula_time`$_{\text{utterance}}$, *unless* the utterance carries `truth_value = performative`. The exemption is grounded in the speech-act tradition of Austin (1962) and Searle (1969): prophecies, vows, oaths, orders, and declarations are not fact-claims about the world but illocutionary acts that posit, commit to, or predict future states. They are accordingly the only utterance class that may legitimately reference an event whose `fabula_time` lies in their own future. The witches' prophecy (`EVT_UTT_PROPHECY_HEATH` in *Macbeth*, `truth_value = performative`, `target_ids = [EVT_MACBETH_CROWNED]`) is the canonical example. The constraint is enforced by `shadow_loom.ingestion._validate_time_ordering` (Rule 5, `severity="error"`) and authored into the social-extraction prompt (`prompts/social_extraction.md` Rule 8). Causal effects *produced* by an utterance are expressed in `causal_topology` as `chain_reaction` edges (EVT_UTT_X → EVT_Y), distinct from referential targets in `target_ids`.

A.3 Ego-graph slicing and AMWN sandbox

Given a query with focal entities F and time anchor t, the ego-graph extractor returns

$$\mathcal{G}_{F,t} = \big(\textsc{Neighbours}_k(F) \cap \textsc{Reconstruct}(\cdot, t)\big),$$

the k-hop spatial / causal / social neighbourhood of F at fabula time t, with belief and trait values reconstructed at t so future information cannot leak into a counterfactual. The `AMWNInstantiator` mirrors $\mathcal{G}_{F,t}$ as a NetworkX `MultiDiGraph` for the simulation sandbox, with edge types {`located_in`, `owned_by`, `causal`, `relationship`, `connected_to`, `communicating_with`, `eavesdropped_by`}.

Causal diagram and AMWN proper. *Separately* from the simulation mirror, `shadow_loom.amwn.build_causal_diagram` strips `WorldStateV1.causal_topology` to a typed structural diagram G and `build_amwn` constructs the *Ancestral Multi-World Network* $G^A(G, W^*)$ of Correa and Bareinboim (2025) over a query's intervention/evidence set $W^* = \{V_t, \ldots\}$. Three implementation details bridge the narrative graph and Definition A.1 of the paper:

1. **Node shadowing.** A counterfactual variable V_t is identified by the pair $(V, \text{proj}(T, \text{An}(V)_{G_{\overline{T}}}))$, where the intervention context T is projected onto V's ancestors in the mutilated diagram. Two copies of V coming from different worlds are *the same node* when their projected contexts agree — this is the AMWN's mechanism for avoiding the exponential k-plet blow-up of Shpitser and Pearl (2007). d-Separation (Rule 2) is then read off G^A via NetworkX `is_d_separator`.

2. **Synthetic relationship-metric nodes.** Relationship metrics (affinity, fear, power_dynamic) live on edges in `social_topology`, not as first-class nodes, so `mutation_social` causal edges would otherwise be invisible to d-separation. We promote each (*src*, *tgt*, *metric*) triple into a synthetic node `REL::`*src*`::`*tgt*`::`*metric* with both endpoint entities and the triggering event as parents. Static-topology axes that were never observed (m.*observed* = `false`) are skipped to avoid injecting phantom variables into latent queries.

3. **Channel and utterance routing.** Communication channels are first-class diagram nodes. Each utterance event is wired *speaker* → *utterance* → *channel* → *addressees*, with bidirectional standing capability edges *channel* ↔ *participant*. A Rule-3 surgery on a channel therefore cuts both the standing capability and every per-utterance content path — exactly the closure property the channel-utterance integration relies on.

Latent-free SCM and the optional U-confounder switch. The diagram is built from the engine's ground-truth `causal_topology`, with no bidirected U arcs; reading G^A for d-separation is therefore **sound** but **incomplete with respect to unobserved confounders** — a latent common cause that LLM extraction missed is silently treated as absent. To make the limitation explicit, the `allow_unobserved_confounders` setting injects, for every pair of distinct nodes that share at least one observed parent, a latent shared parent `U_`*a*`__`*b*; d-separation on the resulting diagram refuses to mark such siblings independent because the latent opens an active path. This produces the sound-but-incomplete behaviour callers expect from the narrative-extraction setting and is recommended for any pipeline where Rule 3 pruning will be promoted from advisory to authoritative.

A.4 Causal physics

The causal physics is a *graphical world model of cause and effect*: a typed dynamics over the shared world graph that implements all three rungs of Pearl's causal hierarchy (Pearl, 2009; Bareinboim et al., 2022): **rung 1 is Correlation** (observation: $P(Y \mid X)$ read off the ego-graph without intervention), **rung 2 is Interventions** ($\mathrm{do}(X = x)$, severing incoming edges into X), and **rung 3 is Counterfactuals** (abduction over latent exogenous values, then a do-operation on the abducted world). Each rung corresponds to one query type in Section 2: `ObservationQuery`, `InterventionQuery`, and `CounterfactualQuery` respectively.

Given a sandbox $\mathcal{G}$ and an intervention specification:

Observation (rung 1 — Correlation): read $P(Y \mid X = x)$ directly off the ego-graph; no edge surgery.

Action (rung 2 — Interventions): $\mathrm{do}(X = x)$ severs every incoming causal edge into X and writes $V_X \leftarrow x$. Channel surgery ($\mathrm{do}(\textsc{ch}.\textit{status} = \text{severed})$) and event invalidation ($\mathrm{do}(e.\textit{truth\_value} = \text{false})$) additionally trigger a *provenance prune*: beliefs whose `acquired_via_event_id` or `acquired_via_channel_id` pointed at the surgically disabled object are dropped from the sandbox before propagation, because they are no longer epistemically licensed in the counterfactual world.

Abduction (rung 3 — Counterfactuals): for evidence E, each entity trait T is updated by a precision-weighted Bayesian blend of the historical sandbox prior and the present-day observed value: writing ι_T for the trait's typed inertia (reinterpreted as the precision of the historical prior) and κ_E for an evidence precision (default 1),

$$T^{\text{post}} = \frac{\iota_T \cdot T^{\text{prior}} + \kappa_E \cdot T^{\text{evidence}}}{\iota_T + \kappa_E}, \tag{1}$$

so high-inertia traits shrink toward the historical baseline and low-inertia traits snap to the evidence. (A legacy inertia-damped variant $T^{\text{post}} = T^{\text{prior}} + (1 - \iota_T)\,(T^{\text{evidence}} - T^{\text{prior}})$ is retained for ablation.) The same Bayes blend is applied per axis to relationship metrics on outgoing social edges. Belief back-propagation is gated by a per-recipient `intelligibility` threshold so a belief that could not plausibly have been acquired through its provenance channel is not reinstated. The do-operation of rung 2 is then re-applied on the abducted world.

Counterfactual (ctf-) calculus pre-flight. Before any sandbox surgery, every Rung-2/3 query is screened against the three rules of the ctf-calculus of Correa and Bareinboim (2025), specialised to a latent-free SCM:

Rule 1 (Consistency) suppresses vacuous interventions $\mathrm{do}(X = x)$ for which the observed factual value of X is already x.

Rule 2 (Independence) builds the AMWN $G^A(G, W^*)$ over the union of evidence and intervention targets, with node-shadowing across worlds whose projected contexts on the ancestral set agree, and tests $(Y_r \perp\!\!\!\perp X_t \mid W^*)$ via d-separation on the AMWN. Evidence flagged as redundant is reported but not silently dropped.

Rule 3 (Exclusion) tests $X \cap \mathrm{An}(Y) = \emptyset$ in the mutilated diagram $G_{\overline{Z}}$ and flags interventions that are provably vacuous with respect to the query targets. The engine ships in *advisory* mode by default (the flagged interventions are surfaced to the auditor but retained in the simulation), since a missed bidirected confounder would otherwise let a substantively meaningful intervention be d-separated into a no-op; *prune* mode is opt-in for confounder-complete topologies.

The construction matches Definition A.1 of Correa and Bareinboim (2025) in spirit, with the following deliberate restrictions: (a) we operate on a latent-free SCM, so soundness for d-separation holds but completeness across worlds with shared unobserved confounders would require explicit bidirected edges that the data model does not yet encode; (b) Rule-2 flags are advisory rather than authoritative; and (c) the relationship-edge axes (affinity, fear, power_dynamic) are lifted into synthetic `REL::`*src*`::`*tgt*`::`*metric* diagram nodes so that `mutation_social` edges contribute to d-separation reasoning.

Propagation. A topological sort over the causal sub-graph (with strongly connected components condensed and the members of each non-trivial SCM cycle blocked with reason `cycle`) applies, for each downstream entity trait T at node u with active parents $\{p_i\}$ and edge weights $\{w_i\}$ (where each $w_i = \sigma_i \cdot \phi_i$ is the product of an evidence-strength weight $\sigma_i \in \{0.25, 0.5, 0.75\}$ and a causal-force scalar ϕ_i, attenuated by a mechanism / world-domain fallback when the edge mechanism is outside the trait's typed `MECHANISM_TRAIT_MAP`):

$$I_i = (V_{p_i} - V_u)\, w_i, \quad \bar{I} = \frac{\sum_i I_i}{\max(1, \sum_i w_i)}, \quad \Delta_u = \begin{cases} \bar{I} - \mathrm{sgn}(\bar{I})\, \iota_u & \text{if } |\bar{I}| > \iota_u + \varepsilon, \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

The $\max(1, \cdot)$ divisor makes the gate scale-invariant only *above unit weight*: a single weak edge with $\sum w_i = 0.1$ is not artificially inflated to clear an arbitrary inertia, while ten stacking weak edges no longer dominate one canonical strong source. A spatial affordance check additionally blocks cross-location influence unless an unbroken `connected_to` path exists. An optional noisy-OR propagation mode replaces the deterministic $|\bar{I}| > \iota_u$ gate with a Bernoulli aggregation of per-edge fire probabilities $p_i = \sigma(\beta\,(|I_i| - \iota_u))$ followed by an $\bigvee_i p_i = 1 - \prod_i (1 - p_i)$ aggregator and a threshold (or, under `execute_distribution`, a Monte-Carlo Bernoulli draw); a post-pass *baseline drift* optionally pulls mutated traits back toward type by $(1 - \iota_u) \cdot \rho$. Both are off by default so the canonical engine is bit-for-bit reproducible. The result is a structured `CausalPhysicsResult{mutations, blocked, hidden_deltas, intervened_nodes, rule3_pruned_interventions, rule2_redundant_evidence, noisy_or_probabilities, pruned_beliefs_count, pruned_utterance_event_ids, disabled_channel_ids}`.

A.5 Narrative physics (affective calculus)

The narrative physics is a *graphical world model of reader affect*: it operates on the same typed graph as the causal physics, but its dynamics scores reader-state functionals (mystery, dramatic irony, suspense, surprise) rather than propagating causal force.

Let A be the anchor pair (t_f, t_s) for fabula and syuzhet cursors, F the focal entities, and write *revealed* for any event e with $s(e) \le t_s$. Define $\mathrm{Anc}(e)$ as the causal ancestors of e in the graph and $B(F, t_f)$ as the union of belief sets of F reconstructed at t_f.

Mystery (Sternberg's curiosity). The Sternberg-style curiosity operationalisation (Sternberg, 1978, 1992; Brewer and Lichtenstein, 1982) aggregates over revealed effects e involving the focal cast (events with $s(e) \le t_s$ and at least one focal participant), pooled with the focal entities themselves as permissible effect-nodes (an entity is itself a causal target of every revealed action against it). Each ancestor's contribution is scaled by three multiplicative factors: a *path-strength geometric decay* along the strongest reverse-path product of edge weights from a to e, depth-capped at $D = 4$ (Trabasso and Sperry, 1985) and computed by single-source Dijkstra on the negated-log-weight reverse graph; a *harm-kind salience* σ_{k_a} from the same σ_k table used by suspense; and a per-effect *curiosity-proximity decay* $\rho_e = \exp(-(t_s - s(e))/\tau_{\mathrm{cur}})$ with $\tau_{\mathrm{cur}} = 8$ syuzhet beats, motivated by Iser's gap theory of reader response (Iser, 1976). Writing $\Pi(a, e)$ for the path-strength weight, the gauge is

$$\mathrm{Mys} = \frac{\sum_{e \in \mathrm{eff}} \rho_e \sum_{a \in \mathrm{Anc}(e),\, s(a) > t_s} \sigma_{k_a} \cdot \Pi(a, e)}{\sum_{e \in \mathrm{eff}} \rho_e \sum_{a \in \mathrm{Anc}(e)} \sigma_{k_a} \cdot \Pi(a, e)}, \tag{3}$$

the salience- and proximity-weighted fraction of focal-effect causes still hidden from the reader. The unweighted ratio $|\{a : s(a) > t_s\}| / |\mathrm{Anc}(e)|$ is the equivalent simple form when all factors collapse to unity. The legacy single-edge fallback for multi-hop ancestors gave a 5-hop weak chain as much weight as a 1-hop strong link — contradicting Trabasso & Sperry's empirical finding that audience traceability decays geometrically with chain depth.

Dramatic irony (Booth, 1974; Muecke, 1969; Gerrig, 1993; Pfister, 1988). Following the classical reader/character knowledge-asymmetry framing, for each focal entity $c \in F$ let K_c be the set of events c knows at the anchor: events c participates in (as actor or target) whose fabula time falls at or before the syuzhet anchor's fabula frontier, plus events referenced in revealed utterances spoken by or addressed to c, plus events c holds a `Belief` about whose provenance still resolves. Each gap event's contribution is then scaled by three additional factors: a *harm-kind salience* σ_{k_e} (tragic irony — a hidden mortal threat the focal does not see — outranks comic irony along the same hierarchy used by suspense and mystery); a *false-belief multiplier* $\phi_e^{(c)} \in \{1, m_{\text{fb}}\}$ when the gap event's actor set intersects the focal's *believed- entity targets* (entities about whom c holds a provenance- valid `Belief`), the canonical Iago $\rightarrow$ Othello / Jacqueline $\rightarrow$ Linnet pattern (Pfister, 1988; Gonzalez, 2024); and a *closure-proximity decay* $\rho_e^{(c)} = \max(\rho_{\min}, \exp(-\Delta_{\text{cls}}/\tau_{\text{iro}}))$ where Δ_{cls} is the syuzhet-index distance to the earliest later position at which c first witnesses an event with $\text{fabula}(e') \geq \text{fabula}(e)$ — the dramatic moment c walks into the scene that exposes the truth (Booth's stable vs. unstable irony (Booth, 1974); defaults $\tau_{\text{iro}} = 6$, $\rho_{\min} = 0.4$). The per-character gap is further scaled by an *action weight* $a_c = \min(\bar{a}, 1 + \alpha \cdot \#\{\text{actor-events by } c \leq t_s\})$ on focal prominence (Pfister, 1988; Sutherland, 2013): a focal driving the plot under false information carries more dramatic charge than a bystander. Writing w_e for event e's narrative intensity (default 1), R for the set of revealed events, and $K = 1$ for the saturation constant,

$$g_c = a_c \cdot \frac{\sum_{e \in R,\, e \notin K_c} w_e \cdot \sigma_{k_e} \cdot \phi_e^{(c)} \cdot \rho_e^{(c)}}{\sum_{e \in R} w_e + K}, \tag{4}$$

$$\text{Iro} = \min\bigl(1,\ \beta \cdot \max_{c \in F} g_c + (1-\beta) \cdot \overline{g_c}\bigr), \qquad \beta = 0.6. \tag{5}$$

The aggregator $\beta \cdot \max + (1-\beta) \cdot \overline{g}$ is Sternberg's single-dominant-gap framing: *one* character's tragic blindness carries the irony charge, with secondary characters as residual contribution. Normalising by the *revealed* mass (rather than total event mass) makes Eq. 5 a Sternberg-style *gap fraction* that can both rise (with new reveals the character has not caught up to) and fall (when participation, addressed utterances, belief acquisition, or closure-proximity discharge close the gap), producing the rise-peak-fall arc the structural-affect literature predicts for canonical irony plots (Booth, 1974; Muecke, 1969): Macduff hearing of his family, Poirot's denouement, Nick's letter to Daisy. Two earlier denominators failed against this criterion. The ratio $\#\text{gaps}/\#\text{revealed}-\text{edges}$ grew its numerator and denominator together and so plateaued at a story-specific asymptote by the third reveal — on Reservoir Dogs it even *decayed* from 0.25 to 0.06 as the protagonist became actor-of-record on more revealed edges. Replacing the denominator with the *full* event mass instead pinned the score into a monotone rise across $21/21$ example-world fixtures, since the denominator stopped moving while the numerator kept growing — contradicting the rise-peak-fall arc the theory predicts.

Suspense (Brewer and Lichtenstein, 1982; Comisky and Bryant, 1982; Ortony et al., 1988; Zillmann, 1996; Lazarus, 1991; Cheong and Young, 2015; Ely et al., 2015). The scorer follows the hope/fear (here *hope/threat*) anticipation pair from the structural-affect and OCC-appraisal traditions — not the information-theoretic uncertainty-reduction formulation of Wilmot and Keller (2020), which is structurally closer to the dramatic-irony scorer above and is borrowed there instead. For each unrevealed event e with strongest incoming causal weight $p_e \in (0, 1]$, we compute a per-event contribution that is then classified as *threat* or *hope* for each focal entity $x \in F$ via the rules below.

Disposition-aware classification (Zillmann, 1996). The legacy actor=hope / target=threat rule is overridden by the social topology: an event whose actor has $\text{aff}(\text{actor} \rightarrow x) \leq -0.2$ is bucketed as a *threat* on x (an antagonist's authored misdeed no longer registers as hope just because they are the actor), and an event whose actor has $\text{aff}(\text{actor} \rightarrow x) \geq +0.2$ propagates as *hope* even when x is its target (rescue propagation: an ally arriving to defuse the threat). Worlds with no populated `social_topology` default the affinity to neutral and fall back to the legacy rule.

Anticipatory proximity weighting (Comisky and Bryant, 1982). Subjective probability of a threat rises with imminence, both temporally and spatially. Each event's contribution is multiplied by a temporal–spatial kernel

$$\mathrm{prox}(e,x) = \exp\Big(-\frac{\Delta t_e^{\mathrm{fab}}}{\tau_t}\Big)\cdot\exp\Big(-\frac{\Delta d_{e,x}^{\mathrm{sp}}}{\tau_s}\Big), \tag{6}$$

where $\Delta t_e^{\mathrm{fab}} = \max(0, t_e^{\mathrm{fab}} - t_{\mathrm{now}}^{\mathrm{fab}})$ is the fabula-time gap from the latest revealed event to the unrevealed event, $\Delta d_{e,x}^{\mathrm{sp}}$ is the shortest-path distance in the spatial topology between e's location and x's, and τ_t auto-scales to the world's median inter-event fabula gap so worlds using `fabula_time_spacing`=1000 and unit-spaced worlds both decay over ~ 6 narrative beats; $\tau_s = 4$ hops.

Persistence multiplier (Brewer and Lichtenstein, 1982). The longer a foreshadowed threat lingers unresolved, the louder it gets. Each event's contribution is also multiplied by $\min(\Pi_{\max},\ 1+\alpha\cdot a_e)$ where a_e is the count of *revealed* causal ancestors of e (proxy for how long the gun has been on the mantle), $\alpha = 0.10$ and $\Pi_{\max} = 1.5$.

Harm-kind salience. Each event is tagged with a salience $\sigma_e \in [0,1]$ inferred from the canonical `mechanism` strings on its incident causal edges, taking the maximum across incident kinds so a stab-in-the-back event wired with both `physical` and `betrayal` edges registers at the higher salience rather than being averaged. The salience ranking follows the appraisal-theory hierarchy of core relational themes (Lazarus, 1991) cross-anchored to OCC's prospect-based emotions (Ortony et al., 1988): existential (1.00) $>$ physical (0.85) $>$ betrayal (0.75) $>$ psychological (0.70) $>$ emotional/relational (0.65) $>$ social/reputational (0.55) $>$ epistemic (0.45). The epistemic floor is held low because the cumulative-mystery scorer (Eq. 3) already covers that surface; suspense should not double-count it. Events with no resolvable mechanism default to `physical` (the modal kind in the corpus).

Aggregating per-entity, per-kind contributions weighted by all of the above:

$$w_{\mathrm{threat}}^{(k)}(x) = \sum_{\substack{e\notin \mathrm{rev}\\ \mathrm{kind}(e)=k\\ \mathrm{bucket}(e,x)=\mathrm{threat}}} \sigma_e\, p_e\, \mathrm{prox}(e,x)\,\pi_e, \tag{7}$$

$$w_{\mathrm{hope}}^{(k)}(x) = \sum_{\substack{e\notin \mathrm{rev}\\ \mathrm{kind}(e)=k\\ \mathrm{bucket}(e,x)=\mathrm{hope}}} \sigma_e\, p_e\, \mathrm{prox}(e,x)\,\pi_e, \tag{8}$$

where $\pi_e = \min(\Pi_{\max},\ 1+\alpha a_e)$ is the persistence multiplier and $\mathrm{bucket}(e,x)$ resolves the threat/hope classification under the disposition rule above.

EFK expected-variance combiner (default). Ely et al. (2015) reformulate suspense as the *expected variance of next-period beliefs* about a terminal outcome — a property of a Bayesian belief martingale rather than a structural-affect heuristic. Adopting this functional form gives suspense the same belief-process shape that our surprise scorer (§16) already has, so the two affective scores become moments of the same underlying process rather than unrelated heuristics. We instantiate this per (focal x, kind k) cell using the revealed threat/hope evidence as a Beta posterior over "next reveal lands threat-side":

$$A^{(x,k)} = \sum_{e\in \mathrm{rev}_{\mathrm{threat}}^{(x,k)}} w_e, \quad B^{(x,k)} = \sum_{e\in \mathrm{rev}_{\mathrm{hope}}^{(x,k)}} w_e, \quad \mu_t^{(x,k)} = \frac{1+A^{(x,k)}}{2+A^{(x,k)}+B^{(x,k)}}. \tag{9}$$

Each unrevealed event e on the cell, with bucket $b_e \in \{\mathrm{threat},\mathrm{hope}\}$, weight w_e and proximity $\rho_e = \mathrm{prox}(e,x)$, defines the Bayes update that *would* occur if e were the next reveal:

$$\mu_e^{+} = \frac{1+A+w_e}{2+A+B+w_e}, \quad \mu_e^{-} = \frac{1+A}{2+A+B+w_e}, \quad \Delta\mu_e = \mu_e^{b_e} - \mu_t, \tag{10}$$

so the realised expected squared belief change, weighted by proximity, is

$$\sigma^2_{\text{fk}} = \sum_{e \in \text{unrev}(x,k)} \frac{\rho_e}{\sum_{e'} \rho_{e'}} (\Delta\mu_e)^2. \tag{11}$$

We normalise by the maximum-suspense reference at the same prior — the squared shift a single composite reveal of total mass $W = \sum_e w_e$ would induce on whichever side moves belief most:

$$\sigma^2_{\max} = \max\Big((\mu^+_W - \mu_t)^2,\ (\mu^-_W - \mu_t)^2\Big), \qquad \widetilde{\sigma}^2_{(x,k)} = \min\Big(1,\ \sigma^2_{\text{fk}}/\sigma^2_{\max}\Big) \in [0,1]. \tag{12}$$

The cell gauges are then aggregated by salience-weighted stakes attenuation:

$$\text{Susp}^{\text{efk}} = \frac{\sum_{(x,k)} \sigma_k \cdot \text{stakes}^{(x,k)} \cdot \widetilde{\sigma}^2_{(x,k)}}{\sum_{(x,k)} \sigma_k \cdot \text{stakes}^{(x,k)}}, \quad \text{stakes}^{(x,k)} = \frac{T^{\text{unrev}}_{(x,k)}}{T^{\text{unrev}}_{(x,k)} + K_k}, \tag{13}$$

where $T^{\text{unrev}}_{(x,k)} = \sum_{e \in \text{unrev}(x,k)} w_e$ is the unrevealed weighted mass on the cell (so stakes decay as the narrative exhausts its forward reveal budget) and K_k is the per-kind saturation constant from the harm-kind table. A *bilateral-mass guard* restricts the aggregator to cells whose unrevealed set contains both threat and hope candidates: a one-sided forward reveal set is despair (only threats coming) or safety (only hopes coming) under Brewer–Lichtenstein structural-affect theory, even though strict EFK would still admit positive variance from magnitude uncertainty alone. This matches the OCC prospect-based-emotion taxonomy (Ortony et al., 1988): suspense requires outcome ambiguity, not merely magnitude ambiguity.

Per-kind balance $\times$ *stakes with weighted-max combine (*`mode=classic`*).* The original Brewer–Lichtenstein-anchored aggregator is retained as an opt-in alternative: rather than collapsing all kinds into one bucket and computing a single balance term — which dilutes a coherent single-kind tension into a multi-kind average — we compute balance and stakes *within* each kind and combine via a salience-weighted max:

$$\text{Susp}^{\text{classic}} = \max_k \Big(\sigma_k \cdot \text{balance}^{(k)} \cdot \text{stakes}^{(k)}\Big), \tag{14}$$

with $\text{balance}^{(k)} = 1 - |w^{(k)}_{\text{threat}} - w^{(k)}_{\text{hope}}|/T_k$, $\text{stakes}^{(k)} = T_k/(T_k + K_k)$, and $T_k = w^{(k)}_{\text{threat}} + w^{(k)}_{\text{hope}}$. This matches Brewer & Lichtenstein's prediction that one *dominant* unresolved beat carries the structural-affect arc, rather than several diffuse anxieties additively. It is also used as the fallback when no kind has both $w^{(k)}_{\text{threat}} > 0$ and $w^{(k)}_{\text{hope}} > 0$. The per-kind saturation constants K_k rise with kind salience so existential threats saturate slowly while social ones saturate fast: $K_{\text{existential}} = 4.0$, $K_{\text{physical}} = 3.0$, $K_{\text{betrayal}} = 2.5$, $K_{\text{psychological}} = 2.0$, $K_{\text{emotional}} = 2.0$, $K_{\text{social}} = 1.5$, $K_{\text{epistemic}} = 1.5$. The dominant kind and per-kind sub-totals are surfaced alongside the scalar so downstream directive consumers can target the dominant beat (*footsteps closing in* vs. *the lie about to surface* vs. *the fellowship about to fracture*). Suspense returns 0 when no kind has both $w^{(k)}_{\text{threat}} > 0$ and $w^{(k)}_{\text{hope}} > 0$ — the suspense–despair and suspense–safety boundaries.

Surprise. Let $\text{actual}_T(x)$ denote entity x's trait T value at the world's terminal fabula time, resolved via $\textsc{Reconstruct}(x, t_{\max})$ so authored `state_timeline` arcs (e.g. Macbeth's ambition rising $0.7 \to 0.85$) are honoured rather than read off the baseline. For each focal entity $x \in F$, the prior q starts at the *leave-one-out* corpus marginal

$$q_0(T, x) = \frac{1}{|\mathcal{N}_{\text{ENT}} \setminus \{x\}|} \sum_{y \in \mathcal{N}_{\text{ENT}} \setminus \{x\}} \text{actual}_T(y), \tag{15}$$

defaulting to 0.5 when fewer than two other entities carry T. Leave-one-out prevents the focal entity from biasing its own prior — with the small casts typical of the example fixtures the inclusive marginal would otherwise pull q toward p and systematically depress KL.

For each revealed cause $(a \to x)$ with weight w the prior is now updated by a *Beta-Bernoulli* step:

$$\alpha \mathrel{+}= w \cdot \text{actual}_T(x), \quad \beta \mathrel{+}= w \cdot (1 - \text{actual}_T(x)), \tag{16}$$

seeded by a Beta$(s \cdot m, s \cdot (1-m))$ prior with weak pseudo-count strength $s = 2$ (strong enough to keep q off the ε-clipped extremes when evidence is sparse, weak enough to remain responsive to the first few edges). The posterior mean $q = \alpha/(\alpha + \beta)$ replaces the legacy geometric pull $q \mathrel{+}= w(\text{actual} - q)$, which was a Storck/Hochreiter/Schmidhuber RDIA proxy with undefined posterior variance. Edges where x is the causal *target* contribute at full weight; edges where x is the causal *source* contribute at $0.4\times$ full weight ("X did Y to Z" speaks more strongly about Z's traits than X's, but X's act is non-trivial evidence about X's traits — Macbeth's ambition is reinforced by acting on it).

Letting $p = \text{actual}_T(x)$,

$$D_{\text{KL}}(p \,\|\, q) = p \log \tfrac{p}{q} + (1-p) \log \tfrac{1-p}{1-q}, \tag{17}$$

the trait-level KL component is the salience-weighted soft- saturated mean

$$\text{Sur}_{\text{KL}} = \frac{\sum_{T \in \mathcal{T}} \sigma_T \big(1 - e^{-D_{\text{KL}}(p \,\|\, q)}\big)}{\sum_{T \in \mathcal{T}} \sigma_T}, \tag{18}$$

where σ_T is the per-trait *narrative salience* (Reagan et al., 2016; arc-relevance hierarchy with ambition / guilt / vengeance / despair / love / loyalty / courage at $\sigma_T \in [0.85, 1.0]$, peripheral traits like literacy / fitness / wealth at 0.30). The saturation constant $\tau = 1$ in $1 - e^{-\text{KL}}$ matches the binary- distribution-discrimination JND from psychophysics (Lu and Dosher, 2013; subjective just-noticeable belief shift in the $[0.5, 1.0]$ nat band). The earlier $\overline{\text{KL}}/\log(1/\varepsilon)$ form divided by the *theoretical* binary-KL maximum ($\log(1/\varepsilon) \approx 4.6$ at $\varepsilon = 0.01$), squashing the entire perceptual signal into the bottom 4% of the gauge — every example-world plot read as flat ≤ 0.10 even when canonical surprise traits (Macbeth's despair, Macduff's grief, Lady Macbeth's guilt) carried per-trait KLs of 0.27–0.50.

The trait-KL component is then combined with a plan-based *anachrony* component (Bae and Young, 2008; Tobin, 2018; Bissell et al., 2025) that captures the surprise generated by temporal reordering — flashbacks that reframe earlier events, openers in medias res:

$$\text{Sur}_{\text{ana}} = \frac{1}{|\mathcal{E}_t|} \sum_{e \in \mathcal{E}_t} \frac{|\text{rank}_{\text{fab}}(e) - \text{rank}_{\text{syu}}(e)|}{N-1}, \tag{19}$$

where $\mathcal{E}_t$ is the set of relevant focal events (cumulative mode: all revealed at t_s; local mode: only those newly revealed at t_s, mirroring the per-step Itti-Baldi spike). The aggregator is a convex split

$$\text{Sur} = w_t \cdot \text{Sur}_{\text{KL}} + w_a \cdot \text{Sur}_{\text{ana}}, \qquad w_t = 0.7,\ w_a = 0.3, \tag{20}$$

which preserves the existing test-contract dominance of trait shifts on linear tellings (anachrony $\to 0$) while letting the non-linear example fixtures (Reservoir Dogs, Gone Girl, Tinker Tailor) accrue the additional anachrony-driven contribution that Bissell et al. (2025) flag as the most important missing dimension in KL-only narrative-surprise models. With no syuzhet anchor the reader is assumed omniscient and $\text{Sur} = 0$.

Six emotion scorers (`grief, rage, joy, regret, love, fear`) score *closeness* to a per-effect trait target. A shared `_EFFECT_TRAITS` table maps each effect to a pair (P_e, I_e) of *positive* and *inverse* trait indicators, calibrated against the actual trait vocabulary of the example-world corpus. Letting $v(x, T)$ denote the current value of trait T on entity x (resolved via RECONSTRUCT), the per-entity score is

$$\text{Emo}_e(x) = \frac{\sum_{T \in P_e \cap \tau(x)} v(x,T) + \sum_{T \in I_e \cap \tau(x)} \big(1 - v(x,T)\big)}{|(P_e \cup I_e) \cap \tau(x)|}, \tag{21}$$

where $\tau(x)$ are the trait names x carries. Both lists drive scoring symmetrically: an earlier filter applied P_e only, leaving I_e as dead code (a brave character routed through `fear` got no fear-reducing credit because `courage` never entered the average). When the intersection is empty — a deliberately traitless cast for that effect — the score falls back to the worst-case $+1$ rather than silently scoring at the midpoint. The combined affective score requested by a `DirectiveQuery` is a weighted sum over the active functionals.

Two-axis sampling and dual-mode surprise. The same scorers are evaluated along the syuzhet axis (advancing t_s only reveals more, so mystery is non-increasing and dramatic irony rises with reveals and falls with character knowledge acquisition) and the fabula axis (snapshotting the world at t_f, where intrinsically surprising story moments can spike). Surprise is exposed in two formally distinct modes corresponding to the two quantities Itti and Baldi (2009) distinguish. The *cumulative* form (Eq. 18, used by the directive-assembly optimiser) is the integrated gap $D_{\mathrm{KL}}(p \,\|\, q_s)$ between the reader's accumulated prior q_s and the truth p, and is monotonically non-increasing on the syuzhet axis. The *local* form (used by the time-series view) is the per-step belief-update $D_{\mathrm{KL}}(q_s \,\|\, q_{s-1})$ — the canonical Itti-Baldi *Bayesian Surprise* definition (the KL distance between the reader's prior immediately after and immediately before the current syuzhet anchor's revelations); the iLab Bayesian-Surprise project page acknowledges this is formally equivalent to the Storck et al. (1995) reinforcement-driven information acquisition formulation (KL divergence between the agent's belief states $p^*_{ijk}(t+1)$ and $p^*_{ijk}(t)$ before and after a state transition). Quiet stretches contribute ~ 0; revelations spike in proportion to how much they shift the prior, producing the impulse-and-decay trajectory the theory predicts and consistent with the corpus emotional-arc trajectories of Reagan et al. (2016) and Kim et al. (2017). Conflating the two forms is a recurring source of misinterpretation: cumulative surprise is a state quantity ("how much catching-up the reader still has to do"), while local surprise is its temporal derivative ("which revelation moments most shift the reader's beliefs").

Propositional belief-state affect (`affect_unification.py`). The four scorers above operate on *trait* and *event* mass — graph geometry, calibrated to behave well as a directive-assembly loss. Alongside them, the post-2026 ingestion pipeline lays down a typed `Proposition` registry (with `audience_default_prior` $\in [0,1]$, `stakes` $\in [0,1]$, time-indexed `truth_at_fabula`, and a sparse `state_timeline`) and per-entity `Concern` rows (polarity desire/fear, salience, activation window, counter-concern pairs). On top of those, `shadow_loom/affect_unification.py` exposes a second, *propositional / Bayesian* scoring layer that operates directly on belief revision over named claims rather than on graph geometry. Letting $p_{\mathrm{aud}}(P, t_f)$ denote the audience belief on proposition P at fabula time t_f (reconstructed from a synthesised `ENT_AUDIENCE` entity's belief replay, falling back to P's `audience_default_prior` when no belief exists) and $p_c(P, t_f)$ the analogous belief of focal character c:

$$\mathrm{Susp}^{\mathrm{prop}}(t_f) = \sum_{P \in \text{open outcomes}} H\big(p_{\mathrm{aud}}(P, t_f)\big) \cdot \mathrm{stakes}(P, t_f) \cdot e^{-\Delta t_f/\tau}, \tag{22}$$

$$\mathrm{Sur}^{\mathrm{prop}}(t_f \,\|\, t'_f) = \sum_{P} D_{\mathrm{KL}}\big(p_{\mathrm{aud}}(P, t_f) \,\|\, p_{\mathrm{aud}}(P, t'_f)\big) \cdot \mathrm{stakes}(P, t_f), \tag{23}$$

$$\mathrm{Iro}^{\mathrm{prop}}(c, t_f) = \sum_{P} D_{\mathrm{KL}}\big(p_{\mathrm{aud}}(P, t_f) \,\|\, p_c(P, t_f)\big) \cdot \mathrm{stakes}(P, t_f), \tag{24}$$

$$\mathrm{Mys}^{\mathrm{prop}}(t_f) = \sum_{e \in \text{known effects}} H\big(\mathrm{softmax}_a \, \mathrm{path}(a, e)\big), \tag{25}$$

where H is binary Shannon entropy, the suspense sum runs over propositions with `kind='outcome'` whose `truth_at_fabula` has not yet committed, the kernel $e^{-\Delta t_f/\tau}$ is the Comisky–Bryant imminence factor on the fabula axis with τ defaulting to the world's median inter-event fabula gap, the surprise form is Itti and Baldi's definition lifted to the proposition layer (and formally equivalent to Storck et al.'s belief-state KL), and the mystery form is Carroll's erotetic mystery: softmax-normalised candidate-cause path strengths in the proposition graph induced by `EventNode.resolves_proposition_ids` and `asserts/denies_proposition_id`, summed over effect propositions whose audience confidence already exceeds the known-effect threshold (default 0.7). The trait layer (above) and the propositional layer coexist: the directive-assembly optimiser consumes the trait scorers because their graph-geometry loss is defined for every world; the propositional layer is what surfaces in the affective dashboard for any consumer asking the Bayesian-narratology question *"how surprised should the reader have been by this beat, given the stated audience prior on this proposition?"*, which is undefined for worlds whose ingestion never produced a Phase A3 proposition catalogue. Valence-decomposed variants are also exposed: `compute_surprise_breakdown` returns a Tan/Ortony pleasant/unpleasant split per focal entity (using the entity's `Concern` polarity to decide whether a positive belief shift on a desired proposition is pleasant surprise vs. pleasant relief), and `compute_irony_breakdown` returns Sternberg's three irony modes (suspense-irony, curiosity-irony, surprise-irony).

Theory-driven refinements (May 2026). A re-reading of the primary sources against corpus output identified five gaps closed by small, interpretable factors on top of the scorers above; each is anchored to a specific literature claim rather than a calibration heuristic.

(i) Audience-concern weighting on suspense and dramatic irony (Tan, 1996). Brewer–Lichtenstein establishes *whether* a future outcome is suspenseful (outcome ambiguity); Tan's *F-emotions* supply *which* outcomes the audience cares about. Without a concern gate, two non-focal NPCs arguing weighted as much as the protagonist on trial. We introduce a per-event multiplier

$$\omega_{\text{con}}(e) = 0.2 + 0.8 \cdot \frac{S_e}{1 + S_e}, \qquad S_e = \sum_{c \in C_{\text{aud}} : \text{ref}(c) \cap \text{part}(e) \neq \emptyset} s_c, \tag{26}$$

where C_{aud} is the audience's concern set, $\text{ref}(c)$ the referent set of the proposition behind concern c, $\text{part}(e)$ the actor/target set of e, and s_c the concern salience. The 0.2 floor avoids zero-weighting orphan events; the saturation maps a single high-salience concern to the ceiling. ω_{con} multiplies w_e in both Eq. 5 (gap mass) and the EFK / classic suspense ledgers (Eq. 13), making both surface and gap forms of structural affect uniformly F-emotion gated.

(ii) Carroll erotetic recency on mystery (Carroll, 1990). Carroll's erotetic narrative theory holds that the mystery salient at any reader position is the set of *live* questions (those raised by recent surface events). The curiosity-proximity decay $\rho_e = \exp(-(t_s - s(e))/\tau_{\text{cur}})$ in Eq. 3 already implements this directly; the May-2026 audit confirmed the alignment and re-anchored the docstring to Carroll's primary claim rather than the previous Iser citation alone.

(iii) False-lead mystery (Sayers, 1929; Knox, 1946). Detective-fiction theory (Sayers' *Omnibus of Crime* introduction; Knox's *Decalogue*) frames mystery intensity as tracking not only the number of unrevealed causes but the number of *plausible-but-wrong* hypotheses the reader is actively entertaining. Red-herring-rich worlds (Christie, *Gone Girl*) read as more mysterious than worlds with the same hidden-ancestor count but no plausible alternates. Approximating the live-hypothesis count as

$$n_{\text{fl}}(t_s) = \big|\{P \in \text{open} : 0.3 \le p_{\text{aud}}(P, t_f(t_s)) \le 0.7\}\big|, \tag{27}$$

the gauge of Eq. 3 is blended with a saturating false-lead term:

$$\text{Mys}^* = 0.75 \cdot \text{Mys} + 0.25 \cdot \frac{n_{\text{fl}}}{n_{\text{fl}} + 4}. \tag{28}$$

(iv) Correlation-aware surprise aggregator (Friston, 2010). Friston's free-energy / predictive-coding framework treats surprise as the precision-weighted prediction error on a single information unit. The naive aggregator double-counts when simultaneously- revealed propositions are causally connected — the audience perceives the joint reveal as a single update, not several. We deflate each per-step KL contribution by

$$w_P = \frac{1}{1 + \kappa \cdot n_{\text{kin}}(P)}, \qquad \kappa = 0.5, \tag{29}$$

where $n_{\text{kin}}(P)$ counts P's causal-graph neighbours that *also* moved this step. One co-moving kin halves the contribution, three quarters it. Applied inside the local form of Sur_{KL} where joint-reveal double-counting is the dominant systematic bias.

(v) Composite narrative tension (Brewer and Lichtenstein, 1982; Sternberg, 1978; Vorderer et al., 1996). The four scorers measure orthogonal facets of structural affect, but readers experience a single felt *tension*, the quantity Brewer & Lichtenstein originally framed as a **triad** (suspense + curiosity + surprise). Sternberg frames the same construct as the *disequilibrium* between what the reader knows, suspects, and is owed; Vorderer et al. (1996) as the running integral of moment-to-moment uncertainty. We aggregate

$$T = 0.40 \cdot \mathrm{Susp} + 0.25 \cdot \mathrm{Mys}^* + 0.20 \cdot \mathrm{Iro} + 0.10 \cdot \dot{\mathrm{Sur}} + 0.05 \cdot D_{\mathrm{unp}}, \tag{30}$$

with the surprise term as its first derivative $\dot{\mathrm{Sur}}$ (the local form: the disequilibrium kick), and $D_{\mathrm{unp}} = n_{\mathrm{unp}}/(n_{\mathrm{unp}} + 4)$ counting foreshadowing arcs whose payoff event has not yet been revealed at the anchor (Chekhov's-gun overhang). The weight vector is calibrated against the `example_worlds/` corpus so canonical mid-arc tension peaks land in the $[0.5, 0.7]$ band; on *Macbeth* at the Duncan-murder anchor $T \approx 0.49$, rising to 0.51 at the Banquo banquet and decaying toward 0.28 at the denouement. Implemented as `DirectiveAssembler.compute_tension_score` and surfaced as the `narrative_tension` series in the affective-curve plot, gauge strip, and `compute_affective_score` dispatch table; also exposed via the MCP server's `target_effect` whitelist and the `DirectiveQuery` schema.

A.6 Generation

`DirectiveAssembler.evaluate_candidate_events()` forks the sandbox per candidate intervention, runs the causal physics, prunes candidates that violate inertia, affordance, or propagation constraints, ranks survivors by the affective score, and packages the winner as a `CreativeBrief` of typed `ConstraintBlocks` (per-effect): hidden-predecessor lists for mystery, "MUST NOT learn" guards for irony, threat/hope tables and hidden-channel lists for suspense, per-trait KL magnitudes for surprise, and per-trait shift constraints with headroom and inertia evidence for the emotion targets. Only the brief is shown to the renderer LLM, which produces dialogue, description, and pacing within the envelope.

A.7 Audit and feedback loop

The auditor (an LLM-as-judge in the sense of Gu et al., 2024) runs a single *categorical* pass per iteration whose prompt is assembled from a target-effect-gated category set (`epistemic` | `probabilistic` | `counterfactual` | `physics` | `meta` | `style`). Three logical check families are bundled into that pass: (i) **causal** — reverse-engineers prose into causal claims and flags "miracle steps" (state changes with no licensing edge in the brief); (ii) **abduction** — runs counterfactual probes against the prose to check that implicit events hold; (iii) **affective** — measures the realised epistemic gap in the prose against the target. Bundling the three families into one prompt rather than three sequential LLM calls trades a small loss of focus for a $3\times$ reduction in LLM cost and round-trip latency, and lets each iteration reason jointly about cross-family interactions (a miracle step is usually *also* an abduction failure). If the structured loss is non-zero, generation is re-run with the auditor's feedback appended, up to `max_correction_retries`. On success the re-extracted prose is merged into a new `VersionRow`, with `world_id` set by the branch policy (`auto` | `mainline` | `shadow`); counterfactual queries default to a fresh `shadow` branch, preserving canon.

Convergence safeguards. The naive iterate-until-the-auditor-passes loop has two failure modes that surface as non-convergence on densely-constrained queries: (a) the rewriter *regresses* — closing the latest violation while reintroducing one that an earlier iteration had already fixed — and (b) the rewriter *flips the rendering mode* (e.g. `counterfactual` → `observation`), so each subsequent audit evaluates a different rubric than the prose was written under. The implementation defends against both: each iteration's violation set $\{(\mathrm{type}, \mathrm{evidence})\}$ is snapshotted, and if iteration $n{+}1$ both *strictly increases* the violation count *and* introduces a previously-unseen violation type, the loop rolls back to the iteration-n prose and exits with a structured `correction_error`; independently, the brief's `rendering_mode` is treated as immutable across the loop, and any rewrite that returns a mismatching mode is rejected as a generation error. Composition- level safeguards run upstream, in the directive assembler: when the brief simultaneously requests a POV-locked perspective *and* a summary source-form (plot_summary, synopsis, outline) the two constraints are mutually unsatisfiable, so the assembler resolves the mutex deterministically (POV wins for epistemic / affective effects that depend on a perspective anchor; summary wins otherwise) and records the decision on the brief so the auditor enforces exactly one. The renderer also drops the *abducted-cue* sub-directive whenever the Rung-3 abduction posterior is flat ($|\Delta| < 0.10$ on every trait), so the prose is no longer asked to dramatise observable behaviour for shifts the simulator itself called negligible. Finally, the ERASED UTTERANCES (HARD) constraint block carries a *structural fingerprint* of each erased speech act (speaker, addressees, target ids, truth-value, channel) rather than the canonical content: showing the verbatim text in a "do not echo this" instruction is the classic pink-elephant anti-pattern that makes the line the most salient phrase in the renderer's context, and the rewriter either reproduces it or paraphrases its evidentiary logic. Deterministic leak detection still reads the canonical content from the world state directly, so removing it from the brief weakens neither the constraint nor the audit.

Cycle handling. The propagator and the ingestion auto-repair pass distinguish *temporal-collapse* cycles from *genuine* causal loops. An affordance_gate edge Entity $\rightarrow$ Event refers to the entity's pre-event state, while a mutation edge Event $\rightarrow$ Entity refers to the post-event state; collapsed onto a single entity node those two classes form a strongly-connected component (SCC) that does not exist in fabula time. Cycle detection therefore runs on the affordance_gate-excluded sub-graph; any SCC that survives is treated as a real defect, and the ingestion repair pass iteratively breaks the lowest-causal_force edge in each surviving SCC until the propagation graph is acyclic (or a configurable iteration cap is hit, which is logged as a warning). Genuine forward-causal loops between events thus remain visible to the auditor while characteristic extraction artefacts (e.g. Vader as both cause and consequence of the duel that mutates him) dissolve without losing causal information.

B Worked examples on real plot fixtures

The repository ships twenty hand-authored example_worlds/ fixtures paired with prose synopses in sample_plots/. A small subset is used below to illustrate each component end-to-end. Field values quoted below are verbatim from the fixtures; queries are the same ones run by the example walkthroughs in docs/model-examples.md and docs/pipeline-by-example.md.

B.1 World model — Macbeth

In example_worlds/macbeth.py, the location LOC_HEATH carries ambient_state{supernatural: AmbientVector(value=0.9, volatility=0.2, evidence_strength="strong"), concealment: 0.7}. The entity ENT_MACBETH is a bundle of typed TraitVectors: ambition, courage (high inertia 0.7, hard to shake), guilt (low inertia 0.25, highly mutable), and so on; an initial Belief(target_id="ENT_WITCHES", perceived_state= "The witches' prophecies may yet prove true", confidence=0.4, inertia=0.35) sits on his belief list. OBJ_BLOODY_DAGGERS declares Affordance(action="kill", target_type="Entity") — the precondition gate the engine checks before it lets EVT_DUNCAN_MURDER fire. Every EventNode carries both indices: the murder has a fabula time but the syuzhet index follows the staged scene. The fixture exercises all five CausalEdge modalities (chain_reaction, mutation, mutation_social, affordance_gate, ambient_propagation); the ambient_propagation edge is what lets LOC_HEATH.supernatural=0.9 bias Macbeth's uptake of the prophecy belief. Beyond Macbeth, the same schema captures wildly different epistemic regimes: CHN_HIDDEN_TELESCREEN_SURVEILLANCE in *Nineteen Eighty-Four* carries intelligibility{ENT_WINSTON: 0.0, ENT_JULIA: 0.0} so every utterance riding it is invisible to the protagonists; CHN_PIP_BENEFACTOR_PIPELINE in *Great Expectations* sets intelligibility{ENT_PIP: 0.1} to encode the whole misattribution arc;

CHN_RHYS_FEYRE_BOND in *ACOTAR* uses 0.4 for partial telepathic bleed. None of these required a schema extension — they are just different fillings of the same Channel.intelligibility map.

B.2 AMWN sandbox — Macbeth "what if Macbeth refuses?"

The query "Macbeth refuses to murder Duncan" is a rung-2 (Intervention) query. The router builds an ego-payload focused on ENT_MACBETH at the Inverness anchor, and AMWNInstantiator.create_sandbox(ego_payload, query_type="intervention") mirrors it as a MultiDiGraph; a sibling VersionRow with world_id="shadow" holds the fork. apply_do_operator({EVT_DUNCAN_MURDER: $\oslash$}) severs the murder's incoming causal edges; the downstream chain_reaction edges into EVT_MALCOLM_FLEES, EVT_MACBETH_CROWNED, EVT_BANQUO_MURDERED all lose activation; the mutation edges into ENT_MACBETH.guilt and paranoia never fire. The factual mainline is untouched — the shadow row is browsable, diffable, and promotable from the UI.

B.3 Causal physics — Macbeth propagation under "Impact $>$ Inertia"

After the do-operator, propagate() topologically sorts the causal sub-graph. For the original (factual) chain, the only active inbound edge into ENT_LADY_MACBETH from EVT_LADY_MACBETH_PERSUADES (a mutation_social edge with trait_target $=$ *power_dynamic* and causal_force $= 7.0$ against ENT_MACBETH) lifts her relational dominance over Macbeth by $\Delta = +0.4$, while the symmetrical edge EVT_BANQUO_GHOST $\rightarrow$ ENT_MACBETH (trait_target $=$ *fear*, trait_delta $= +0.5$, causal_force $= 6.0$ against ENT_BANQUO) shocks Macbeth's *fear-of-Banquo* channel by $\Delta = +0.5$. With a single active source per channel the normaliser $\max(1, \sum w_i) = 1$, so $\bar{I} = I$, and both are above the trait inertia (Lady Macbeth's resolve-related inertia ~ 0.3; Macbeth's per-relation inertia for an unestablished fear is low). A symmetrical attempt to shock Macbeth's courage (initial inertia 0.7) at $w \approx 0.5$, $\Delta_{\text{src}} = 0.4$ would give $\bar{I} = 0.20 < 0.7$ and is *blocked* — exactly the play's psychology, in which his nerve outlasts his conscience. With multiple stacking weak sources the divisor switches to $\sum w_i$ and the aggregate behaves as a weighted average, preventing ten weak rumours from outweighing one canonical strong cause. Spatial affordance gating additionally blocks cross-location shocks unless a connected_to path exists. The truth-value guard generalises the same gate to utterance evidence: EVT_UTT_NOELLE_PREGNANCY_CLAIM (Gone Girl, truth_value="false") and EVT_UTT_BALTHASAR_REPORTS_JULIETS_DEATH (Romeo and Juliet, truth_value="false") both fire mutation_social edges, so the addressees update their beliefs about *what was said* and the social aftermath plays out, but the abduction step refuses to use the falsehoods as factual support; downstream beliefs whose acquired_via_event_id cites those utterances therefore carry the deception in their provenance and are pruned automatically if the originating utterance is severed by a counterfactual.

B.4 Pearl-rung worked numerics — Macbeth (real engine output)

The numbers below are the actual CausalPhysicsResult.mutations, hidden_deltas and rule3_pruned_interventions returned by CausalPhysicsEngine.execute() on the bundled example_worlds/macbeth.py fixture, focal cast {Macbeth, Lady Macbeth, Duncan, Banquo, Macduff}. Reproducible via scripts/_dump_pearl_rungs.py.

Rung 1 — Observation (interventions $= \emptyset$). Forward propagation of ambient sources only: three trait shifts pass the noisy-OR gate (ENT_BANQUO.suspicion: $0.003 \rightarrow 0.015$; ENT_MALCOLM.leadership: $0.204 \rightarrow 0.214$; ENT_LADY_MACBETH.ruthlessness: $0.993 \rightarrow 0.999$), and 27 weaker impulses are absorbed (result.blocked, reason="noisy_or_absorbed").

Rung 2 — Intervention (do(ENT_MACBETH.ambition$=0$), target set {ENT_DUNCAN, ENT_LADY_MACBETH}). intervened_nodes $=$ {ENT_MACBETH}, rule3_pruned_interventions $= \emptyset$. The do-surgery severs the inbound edges into ambition and pins the trait at 0; sibling traits remain free. Seven trait mutations fire on top of the pin:

Node	Trait	old	new	impact
ENT_LADY_MACBETH	resolve	+0.966	+0.980	+0.040
ENT_LADY_MACBETH	ruthlessness	+0.698	+0.704	+0.019
ENT_LADY_MACBETH	guilt	+0.922	+0.917	−0.006
ENT_LENNOX	caution	+0.862	+0.868	+0.023
ENT_LENNOX	suspicion	+0.242	+0.253	+0.023
ENT_BANQUO	suspicion	+0.001	+0.013	+0.022
ENT_MALCOLM	courage	+0.713	+0.722	+0.026

A vacuous variant do(ENT_DUNCAN.kindness=0) targeted at ENT_BANQUO still produces five propagation mutations in advisory mode (e.g. ENT_LADY_MACBETH.guilt: $0.015 \to 0.063$, ENT_LENNOX.loyalty: $0.830 \to 0.855$); under rule3_pruning_mode="prune" the engine would short-circuit because no directed path survives the mutilation.

Rung 3 — Counterfactual. Same do, with evidence_node_ids = {Macbeth, Lady Macbeth}. Abduction populates result.hidden_deltas with the per-trait latent shifts the Bayesian blend (Eq. 1) requires:

Node	Trait	Δ
ENT_MACBETH	ambition	+0.598
ENT_MACBETH	courage	−0.061
ENT_MACBETH	loyalty	+0.102
ENT_MACBETH	guilt	+0.444
ENT_MACBETH	paranoia	+0.173
ENT_MACBETH	ruthlessness	−0.289
ENT_MACBETH	despair	+0.670
ENT_LADY_MACBETH	ambition	−0.076
ENT_LADY_MACBETH	ruthlessness	+0.188
ENT_LADY_MACBETH	resolve	−0.800
ENT_LADY_MACBETH	guilt	+0.950

The $+0.598$ ambition shift on Macbeth and the -0.800 resolve shift on Lady Macbeth are precisely the latent perturbations the observed Act-V evidence requires given the per-trait inertia and the gap between the sandbox prior and the factual state_timeline. Forward propagation then fires five additional mutations on top of those staged sources (ENT_LADY_MACBETH.guilt: $0.792 \to 0.841$; ENT_LADY_MACBETH.resolve: $0.484 \to 0.495$; ENT_DUNCAN.trust: $0.870 \to 0.875$, etc.). rule3_pruned_interventions flags the do as vacuous on the world-cropped diagram (the over-strict closed-world behaviour discussed in §3.2), but advisory mode keeps it in the simulation so the abduction-driven downstream still mutates.

Romeo and Juliet — do(ENT_FRIAR_LAURENCE.diligence=1). The same engine on the bundled fixture produces 17 trait mutations across the supporting cast — including ENT_BALTHASAR.loyalty: $0.851 \to 0.946$ (impact $+0.233$) and ENT_MERCUTIO.loyalty: $0.988 \to 1.000$ (impact $+0.175$) — demonstrating that a counterfactually diligent friar shifts the supporting cast's allegiance vectors well beyond Romeo and Juliet themselves.

Gone Girl — abduction with no surviving propagation. The Rung-3 query do(ENT_AMY.deceit=0) conditioned on {ENT_NICK, ENT_AMY} produces substantial hidden_deltas (e.g. ENT_NICK.adaptability: $+0.839$; ENT_NICK.resentment: $+0.640$; ENT_AMY.narcissism: -0.196; ENT_AMY.manipulation: -0.100) and *zero* forward mutations: the abduction fully explains the observed Nick/Amy state without any post-hoc forward propagation needing to fire. This is the engine reporting that the do-surgery plus evidence is *consistent* with the observed downstream — the strongest Rung-3 outcome shape.

B.5 Narrative physics — Death on the Nile, Reservoir Dogs, Romeo and Juliet, Macbeth

The four structural scorers each have a canonical fixture.

Mystery (Macbeth Act II). Eq. 3 on the revealed effect `EVT_DUNCAN_DISCOVERED_MURDERED`: of its causal ancestors, `EVT_LADY_MACBETH_PERSUADES` and `EVT_SERVANTS_FRAMED` are still hidden at the audience's `syuzhet_anchor` early in the act; the score is $2/3 \approx 0.67$, falling to 0 once Act V resolves the provenance. The full scorer trajectory across the bundled `example_worlds/macbeth.py` fixture (12 evenly-spaced syuzhet anchors, focal cast Macbeth, Lady Macbeth, Macduff, Duncan, Malcolm, Witches) traces the canonical Sternberg collapse: $0.99, 0.93, 0.87, 0.73, 0.66, 0.61, 0.59, 0.55, 0.41, 0.37, 0.28, 0.28$ — starting near 1 (every ancestor of every revealed effect is still hidden), collapsing through Acts II–III as Banquo's death, the banquet ghost, and the witches' second prophecy reveal earlier hidden causes, and ending at 0.28 once Macduff's family slaughter and the moving forest have closed most of the structural gaps. The same scorer on Death on the Nile traces $1.00 \to 0.29$, on Wuthering Heights $0.99 \to 0.18$, on Tinker Tailor $0.98 \to 0.20$.

Dramatic irony (Death on the Nile). The reader knows Jacqueline and Simon are conspirators (`EVT_PLAN_HATCHED` revealed at low syuzhet); Poirot and the suspect pool do not. With $F = \{\text{ENT\_PENNINGTON}, \text{ENT\_VAN\_SCHUYLER}, \text{ENT\_TIM}\}$, the conspiracy and its downstream revealed effects sit in the $\sum_{e \in \text{rev}, e \notin K_c} w_e$ numerator of Eq. 5 for every $c \in F$. The gauge climbs through the cruise scenes as Poirot's reveals accumulate without reaching the suspect pool, and falls only at the denouement when Poirot briefs the cast and the events enter every K_c. Run on the bundled fixture with the top-6 focal entities (Simon, Jacqueline, Linnet, Poirot, Race, Richetti), the curve is $0.14, 0.30, 0.26, 0.40, 0.48, \mathbf{0.58}, 0.47, 0.42, 0.37, 0.50, 0.28$, peaking at anchor 16 (Poirot's confrontation in the lounge). The same scorer on Macbeth peaks 0.63 at anchor 25 (Macduff's discovery of his murdered family); on Romeo and Juliet at 0.72 around anchor 25 (the crypt-misreading sequence); on Tinker Tailor Soldier Spy at 0.73 at anchor 41 (the cusp of the mole reveal). 16 of 21 bundled fixtures produce a clean rise-peak-fall.

Suspense (Romeo and Juliet, tomb scene). At the tomb $F = \{\text{ENT\_ROMEO}\}$, unrevealed events with Romeo as non-acting target (Friar's letter intercepted, Juliet's draught about to wear off) sum to w_{threat}; events Romeo himself authors and that have not yet fired (drink poison, kill Paris) sum to w_{hope}. Threat dominates. As Romeo drinks, $w_{\text{hope}} \to 0$ and the score returns 0 — the suspense–despair boundary built into the balance $\times$ stakes combiner (Eq. 5 ff.) is hit cleanly. The full curve across the bundled fixture is $[0.14, 0.15, 0.10, 0.09, 0.14, \mathbf{0.24}, 0.12, 0.17, 0.17, 0.00]$ at 10 anchors, peaking at anchor 21 (the moment between Juliet's potion and Balthasar's report). The same gauge on Gone Girl peaks 0.31 at anchor 19 (Amy's mid-novel re-emergence); on Tinker Tailor at 0.39 on the cusp of the mole reveal; on Death on the Nile at 0.28 on Poirot's late-night confrontation. Every world's terminal anchor falls to 0.00 — Wilmot's safety condition, no unrevealed threats remain post-resolution.

Surprise (Reservoir Dogs reveal). `EVT_ORANGE_REVEALED` sits at high fabula time ($f \approx 9000$) but earlier syuzhet ($s = 11$ relative to the heist's chronological ordering, but late in the audience's presentation), with the underlying ground truth (`ENT_ORANGE.constants = ["undercover_cop", ...]`) present at fabula time zero. Eq. 16 initialises a Beta-Bernoulli prior over Orange's loyalty trait at $\text{Beta}(s \cdot m,\ s \cdot (1-m))$ with pseudo-count strength $s = 2$ and corpus-marginal mean $m \approx 0.5$; the few revealed edges into Orange before the reveal (`EVT_ORANGE_KILLS_BLONDE` at full target weight, the earlier paternal-tenderness edges from `ENT_WHITE` at the source-side weight 0.4) keep the posterior mean $q \approx 0.6$ "criminal". After the reveal the actual value $p \approx 0.95$ "cop", producing a per-trait D_{KL} of ≈ 0.6 nat which the soft-saturation $1 - e^{-\text{KL}}$ in Eq. 18 maps to a trait-KL component of ≈ 0.45. The fixture's in-medias-res structure (the heist appears as a flashback after the warehouse aftermath) drives the anachrony component to ≈ 0.30, and the convex aggregator $0.7 \cdot \text{Sur}_{\text{KL}} + 0.3 \cdot \text{Sur}_{\text{ana}}$ yields the corpus-audit peak surprise of ≈ 0.19 at the reveal anchor (see Tab. 2). The reveal is large but *licensed* — the engine had the cause present at low fabula time, only locked behind syuzhet, so re-extraction will not flag a miracle step.

Emotion targets (Gone Girl, Brief Encounter). The six emotion scorers (`grief, rage, joy, regret, love, fear`) resolve their effect through the shared `_EFFECT_TRAITS` table (Eq. 21).

On Gone Girl, the directive "maximise grief in Nick" is scored against grief's positive indicators (`despair, shame, longing, grief, anguish, mourning`) and inverse indicators (`hope, joy, contentment, satisfaction`) intersected with Nick's actual trait set, with the available headroom against those targets determining whether the directive is even feasible at the chosen anchor. On Brief Encounter the same machinery realises `regret` as the gap between Laura's current `self_actualisation` and the regret-positive trait values `guilt, remorse, despair` under a constant-supervision channel that never permits the choice to be made.

B.6 Directive assembly — Romeo and Juliet

The directive "raise dramatic irony to 0.85 in the tomb scene without giving Romeo correct information" triggers `DirectiveAssembler.evaluate_candidate_events()`. From unblocked affordances and reachable spatial paths it enumerates candidates — *Friar's letter intercepted*, *Balthasar arrives faster*, *Juliet wakes one minute earlier* — and forks an AMWN sandbox per candidate. The plausibility gate drops *Balthasar arrives faster* (the plague quarantine `SpatialEdge.is_locked` blocks the path); the remaining sandboxes are propagated, and the survivors are scored. *Friar's letter intercepted* wins because it raises Eq. 3's sister gap-count for irony without adding any utterance event into $B(\text{ENT\_ROMEO}, t_f)$. The winner is wrapped in typed `ConstraintBlocks`: a `must_events` list, a `must_not_events` list (Romeo learning the truth), trait envelopes (Romeo's `despair` envelope), and a syuzhet-window constraint. That bundle is the *only* thing handed to the renderer.

B.7 Generation — the brief as safety envelope

The renderer LLM is given the Romeo-and-Juliet brief as system prompt plus a small style context. It may write any prose about the tomb, the messenger, the dagger; it may not invent a causal edge, shift a trait outside its envelope, open a channel that does not exist, or place an utterance the brief does not licence. The output is a scene of dialogue, description, and pacing inside the mathematical envelope chosen by directive assembly — a typed, causal-graph-conditioned instance of the plan-and-render pattern (Riedl and Young, 2010; Yao et al., 2019; Goldfarb-Tarrant et al., 2020; Yang et al., 2023).

B.8 Corpus-scale audit of the four structural scorers

The four structural-affect scorers — mystery (Eq. 3), dramatic irony (Eq. 5), suspense (Eq. 13), and surprise (Eq. 16–18) — were audited end-to-end against the bundled twenty-fixture corpus in `example_worlds/`. Each fixture was sampled at seven evenly-spaced syuzhet anchors, giving 162 score evaluations per metric (a single-anchor terminal-flat fixture contributes only six points). The audit script `scripts/audit_affective.py` reproduces the table.

Scorer	min	median	mean	max
`mystery`	0.17	0.53	0.57	1.00
`dramatic_irony`	0.00	0.45	0.45	0.90
`suspense`	0.00	0.16	0.14	0.39
`surprise` (local)	0.00	0.03	0.05	0.24
`grief`	0.00	0.12	0.18	0.80
`rage`	0.00	0.48	0.42	0.90
`joy`	0.00	0.76	0.60	1.00
`regret`	0.00	0.10	0.16	0.78
`love`	0.00	0.50	0.43	0.95
`fear`	0.00	0.34	0.28	0.63

Table 2: Per-scorer scale summary across 20 fixtures $\times$ 7 syuzhet anchors. The four structural scorers occupy different absolute bands by design: mystery is a population fraction near 1 early in the syuzhet; surprise is a per-step KL spike near 0 between revelation points. The six emotion scorers measure per-entity *closeness* to a per-effect trait target and so are flat across syuzhet but vary across worlds and entities.

The audit confirms the canonical signatures the underlying theory predicts. *Mystery* falls monotonically across every world (Trabasso & Sperry causal-resolution: as causes are revealed, the hidden-ancestor fraction collapses); on Macbeth the curve is $0.99 \to 0.28$, on Death on the Nile $1.00 \to 0.29$, on Wuthering Heights $0.99 \to 0.18$. *Dramatic irony* produces the rise-peak-fall arc Sternberg's gap-fraction analysis predicts in $16/20$ worlds: Macbeth peaks at anchor 5 (0.62) on Macduff's discovery of his murdered family; Death on the Nile peaks at anchor 4 (0.65) on Poirot's withheld knowledge; Romeo and Juliet peaks at anchor 5 (0.75) on the crypt-misreading sequence. *Suspense* discharges to 0 at the terminal anchor of every world (no unrevealed threats remain post-resolution); the highest mid-story peak is Tinker Tailor (0.39) on the cusp of the mole reveal. *Surprise* spikes sparsely at canonical revelation points: Reservoir Dogs Mr Orange flashback (0.19, dominated by anachrony), Gone Girl mid-novel perspective shift (0.20, dominated by Beta-Bernoulli posterior collapse on Amy's loyalty trait), Wuthering Heights in-medias-res frame opening (0.24, anachrony spike from Lockwood's retrospective frame).

B.9 Tunables (`DirectiveAssemblySettings`)

All scorer constants are externalised through a single `DirectiveAssemblySettings` object (Pydantic `BaseSettings`, env-var prefix `DIRECTIVE_ASSEMBLY_*`), enumerated below with their calibrated defaults. The defaults reproduce the rise-peak-fall arcs in Tab. 2; override individual constants only when targeting non-canonical genres (e.g. raise `IRONY_AGGREGATOR_BETA` toward 1 for ensemble-cast tragedies where a single dominant blindness should overwhelm the average; lower `SUSPENSE_PROXIMITY_TAU_FABULA_GAPS` for thriller pacing where threat sharpens with proximity faster than the corpus default).

Setting	Default
Mystery (Trabasso & Sperry; Iser)	
`MYSTERY_PATH_DECAY_DEPTH`	4
`MYSTERY_PROXIMITY_TAU_SYUZHET`	8.0
Dramatic irony (Booth; Pfister; Cabanas)	
`IRONY_SURFACE_K`	1.0
`IRONY_FALSE_BELIEF_MULT`	1.5
`IRONY_ACTION_ALPHA`	0.15
`IRONY_ACTION_WEIGHT_CAP`	3.0
`IRONY_AGGREGATOR_BETA`	0.6
`IRONY_PROXIMITY_TAU_SYUZHET`	6.0
`IRONY_PROXIMITY_FLOOR`	0.4
Suspense (Lazarus; OCC; Brewer)	
`SUSPENSE_STAKES_K`	2.0
`SUSPENSE_PROXIMITY_TAU_FABULA_GAPS`	6.0
`SUSPENSE_PROXIMITY_TAU_SPATIAL`	4.0
`SUSPENSE_PERSISTENCE_ALPHA`	0.10
`SUSPENSE_PERSISTENCE_CAP`	1.5
`SUSPENSE_HOSTILE_AFFINITY`	−0.2
`SUSPENSE_ALLY_AFFINITY`	0.2
Surprise (Bae & Young; Reagan; Bissell-Paulin-Piper)	
`SURPRISE_TRAIT_KL_WEIGHT`	0.7
`SURPRISE_ANACHRONY_WEIGHT`	0.3
`SURPRISE_DEFAULT_TRAIT_SALIENCE`	0.55
`SURPRISE_SOURCE_EDGE_WEIGHT`	0.4
`SURPRISE_PRIOR_PSEUDOCOUNT`	2.0

Table 3: Externalised affective-scorer tunables (env-var prefix `DIRECTIVE_ASSEMBLY_*`; full table including the `HARM_KIND_SALIENCE` and `TRAIT_NARRATIVE_SALIENCE` dictionaries in `docs/settings.md §8`).

B.10 Audit and feedback loop — Gone Girl

Gone Girl is the system's lying-narrator fixture: Amy's diary is a sequence of `EventNodes` with `truth_value="false"` broadcast through `CHN_DIARY_PUBLIC`. After generation, `auditor.run_feedback_loop` runs a single categorical pass whose prompt bundles the causal, abduction, and affective check families. The *causal* family reverse-engineers the prose into (source, target, modality) claims and matches each against the brief; an LLM-emitted "Nick suddenly realises Amy is alive" with no licensing causal edge is flagged as a *miracle step*. The *abduction audit* runs counterfactual probes ("if the diary were truthful, would the prose still hold?"); a "yes" here means the prose has collapsed Amy's two faces into one, and is rejected. The *affective* family measures the realised dramatic-irony gap against the target. A *style-fidelity audit* compares the prose against the `NarrativeStyle` profile recovered at ingestion — word-count budget, prose density, register, POV, tense, and form class (`news_article`, `historical_account`, `thought_experiment`, etc.) — raising `style_mismatch` on drift. A *meta-narration audit* runs on counterfactual and abduction-driven scenes: it flags prose that comments on its own branch structure (*timeline*, *divergence*, *the alternative holds*, conditional/subjunctive framings used to *describe* rather than *narrate* the counterfactual) instead of rendering the alternate world as a lived past-tense scene, raising `meta_narration`. The aggregated `FeedbackLoopResult` exposes both an `engine_thresholds_passed` gate (deterministic: number of miracle steps, max trait drift) and an `llm_pass` gate (literary critique); either failing re-runs generation with the audit feedback appended, up to `max_correction_retries`. On convergence the scene is re-extracted into a new `VersionRow`; counterfactual branches land on `world_id="shadow"` and only become canon via explicit `promote_branch` (Correa and Bareinboim, 2025).

C End-to-end walkthrough on *Macbeth*

This appendix describes a single complete pass of the pipeline in plain prose. The goal is to make every intermediate object inspectable to a reader who has read the main text but has not read the source. The fixture used throughout is `example_worlds/macbeth.py`, paired with the synopsis in `sample_plots/macbeth.txt`; the query is the canonical counterfactual "what if Macbeth refuses to murder Duncan?". The same loop runs identically on the other nineteen bundled fixtures.

C.1 Step 0: the source text and the user's query

The user opens the workspace, drops the synopsis into the STORY tab, and clicks *Save & Re-ingest*. The synopsis is a five-act prose summary in the public-domain register: the witches on the heath, Duncan's offer of the Cawdor title, Lady Macbeth's persuasion, the murder in Inverness, the banquet, the apparitions, the slaughter at Macduff's castle, the sleepwalking scene, Birnam Wood, and the final duel. After ingestion the user types a query into the REASONING tab: *"Counterfactual: Macbeth refuses to murder Duncan at Inverness. Show what changes for Macduff, Banquo, Malcolm, and the witches, and re-render the banquet scene."* The query is parsed by `query_parsing.parse_query` into a typed `CounterfactualQuery` with focal entities $F = \{\text{ENT\_MACBETH}\}$, target $X = \text{EVT\_DUNCAN\_MURDER}$, intervention value $x = \oslash$ (do-not-occur), and a re-render directive scoped to the banquet's syuzhet window.

C.2 Step 1: ingestion in detail

Ingestion is the only stage where the LLM is allowed to author graph structure rather than prose. It is a fan-out / fan-in pattern over the chunked source.

Ontology pre-pass. Five extraction agents run over the full text in parallel: `ontology_locations` produces `LOC_HEATH`, `LOC_INVERNESS_CASTLE`, `LOC_FORRES_COURT`, and so on, each with an initial `ambient_state` dictionary; `ontology_objects` extracts `OBJ_BLOODY_DAGGERS`, `OBJ_LETTER`, and `OBJ_CROWN` with explicit `Affordance` lists; `ontology_entities` produces `ENT_MACBETH`, `ENT_LADY_MACBETH`, `ENT_DUNCAN`, and the rest, each as an `Entity` with a sparse initial `TraitVector` bundle, beliefs, and a first-appearance `location_id`; `ontology_world_traits` introduces the world-level named latents (`WORLD_AMBITION`, `WORLD_FATE`, `WORLD_GUILT`) that act as common causes; and `ontology_aliases` builds a fuzzy alias table ("Lord of Glamis" $\mapsto$ `ENT_MACBETH`). The result is a populated `GlobalRegister`.

Per-chunk specialist fan-out. The text is chunked into $\sim$ 800-token windows aligned on paragraph boundaries. Each chunk dispatches three concurrent specialist agents that share the `GlobalRegister` as system context: `physics_extraction` produces `EventNodes` with `event_type` drawn from {`choice, outcome, revelation, utterance`}, `causal` edges with `causality_type`, and `spatial` edges (e.g. a `connected_to` edge from `LOC_INVERNESS_CASTLE` to `LOC_FORRES_COURT`); `social_extraction` produces `RelationshipEdges` with per-axis `RelationshipMetrics` for affinity, fear, and power dynamic, and the `Channels` (the witches' prophecy channel, the dinner-table channel at the banquet, the diary-style private channel of Lady Macbeth's letter); and `consequences_extraction` writes the authoritative `state_timeline` entries on each entity, including the inertia bumps that record how shocking events make traits *harder* to dislodge afterwards (Macbeth's `guilt` inertia rises from 0.25 to 0.4 after the murder). The three prompts share a *symmetric per-axis coverage rule*: whenever `social_extraction` emits a `RelationshipMetric` on any of the three axes, `physics_extraction` must emit at least one `mutation_social` edge with the matching `trait_target`; the validation pass enforces the parity. The rule prevents *static dyads* — relationships whose value is non-zero but never causally moved — from producing flat horizontals on the downstream conflict, danger, and power gauges. The bundled corpus (§B) honours this invariant on every observed axis on every dyad.

Validation and auto-repair. The merged graph passes through `_programmatic_validation`, which checks ID closure (every `source_id` resolves), type constraints (every `causality_type` is one of the five enums), basic acyclicity within causal edges of the same fabula slice, and "alive-actor" constraints (no event sourced from a dead character). Failures are passed back to a small repair LLM with the offending field and a structured error; the repair loop runs up to three iterations before raising. Fabula times are then remapped to a uniform $\Delta t = 100$ spacing so that downstream windows are stable across re-ingestion. The output is a single `WorldStateV1`, persisted as the root `VersionRow`.

C.3 Step 2: query routing and ego-graph slicing

The `CounterfactualQuery` is dispatched to the rung-3 (Counterfactual) handler. The handler first consults the `NarrativePhysics` router, which selects an *ego-graph* around the focal entities at the query's fabula anchor: Macbeth's k-hop neighbourhood at $t \approx 6000$ (the Inverness night). The slice contains `ENT_MACBETH`, `ENT_LADY_MACBETH`, `ENT_DUNCAN`, the chamber attendants, `OBJ_BLOODY_DAGGERS`, `LOC_INVERNESS_CASTLE`, the relevant beliefs, and the immediate causal context (`EVT_LADY_PERSUADES`, the prophecy events, the upcoming `EVT_DUNCAN_MURDER`). Trait values are reconstructed at t by replaying `state_timeline` deltas, which prevents post-murder trait shocks from leaking into the counterfactual past.

C.4 Step 3: AMWN sandbox and the do-operator

`AMWNInstantiator.create_sandbox(ego_payload, query_type="counterfactual")` mirrors the ego-graph as a NetworkX `MultiDiGraph`, in the spirit of the Ancestral Multi-World Network construction of Correa and Bareinboim (2025). A sibling `VersionRow` with `world_id="shadow"` is opened to hold the fork. The do-operator `apply_do_operator({EVT_DUNCAN_MURDER: ∅})` severs the murder's incoming causal edges (specifically the `chain_reaction` edge from `EVT_LADY_PERSUADES`, the `affordance_gate` edge from `OBJ_BLOODY_DAGGERS`, and the `ambient_propagation` edge from `LOC_INVERNESS_CASTLE.tension`) and overrides its realisation to "not-occurred". Crucially the factual mainline is untouched. A user looking at the version sidebar sees a green `factual` row with a violet `shadow` child, both browsable.

C.5 Step 4: causal physics propagation

The propagator topologically sorts the causal sub-graph from the intervened node downward and applies Eq. 2 at every edge. Concretely:

- `EVT_DUNCAN_MURDER` $\to$ `EVT_MALCOLM_FLEES` (`chain_reaction`, $w = 0.85$): the source's outcome is "$\emptyset$", so the impact is zero, and `EVT_MALCOLM_FLEES` is marked `blocked_by_intervention`.

- `EVT_DUNCAN_MURDER` → `ENT_MACBETH.guilt` (`mutation`, $w = 0.7$): blocked. The downstream `state_timeline` delta that would have raised guilt from 0.1 to 0.7 is suppressed; the entity's reconstructed trait at the banquet remains close to its battlefield baseline.
- `EVT_DUNCAN_MURDER` → `EVT_BANQUO_SUSPECTS` (`chain_reaction_via_belief`, $w = 0.6$): blocked. `ENT_BANQUO`'s suspicion belief never fires; the `Belief(target=ENT_MACBETH, perceived_state= "Macbeth might be the king's killer")` is absent from his reconstructed belief set in the shadow branch.
- `EVT_LADY_MACBETH_PERSUADES` → `ENT_LADY_MACBETH` (`mutation_social`, axis *power_dynamic* against `ENT_MACBETH`, $\Delta = +0.4$): *not* blocked, because the persuasion event itself is upstream of the intervention. Lady Macbeth's relational dominance over Macbeth still spikes; her subsequent collapse arc is then re-evaluated against the now-absent causal trigger of his guilt.
- Spatial affordance check at every step: a putative shock from `LOC_DUNSINANE_CASTLE` to `ENT_MACDUFF` requires a connected spatial path; if Macduff is in `LOC_ENGLAND` with no open channel, the shock is blocked.

The result is a structured `CausalPhysicsResult{mutations: [...], blocked: [...], hidden_deltas: [...], intervened_nodes: [EVT_DUNCAN_MURDER]}`. The `blocked` list is what makes the engine inspectable: every causal edge that *would have* fired in the factual mainline is recorded with the reason it did not.

C.6 Step 5: narrative-physics scoring

The four structural scorers and the six emotion scorers are now re-evaluated on the shadow branch. The user's directive ("show what changes for Macduff, Banquo, Malcolm, and the witches") biases the dramatic-irony scorer toward those four entities. The counterfactual collapses several gaps that defined the factual mainline: the audience no longer sees Banquo's suspicion gap, no longer sees Macduff's grief-driven vengeance, and the apparition scene is now etiologically dangling (the witches still prophesy, but Macbeth has no kingship to defend). The mystery score on `EVT_DUNCAN_FOUND_DEAD` drops to zero because the event no longer fires; the suspense at the banquet resets, since Banquo's ghost depended on the murder. The directive assembler records this delta as a `CandidateScoreReport` per candidate continuation it considers for the banquet rewrite.

C.7 Step 6: directive assembly and the brief

`DirectiveAssembler.evaluate_candidate_events()` now has to choose what to put in the banquet scene of the shadow branch. Candidate events are enumerated by walking unblocked affordances reachable from the shadow-branch state: *Macbeth confesses the prophecy to Banquo*, *Lady Macbeth persists and stages the murder herself*, *Duncan publicly names Macbeth as the new Prince of Cumberland*, *Macbeth retreats to Glamis to think*. Each is forked into its own micro-sandbox and propagated. Plausibility gates drop *Lady Macbeth stages the murder herself* because the daggers' `Affordance(action="kill")` is gated on a wielder with sufficient `ruthlessness` and the trait-evidence support is low for Lady Macbeth in isolation. The remaining survivors are scored against the directive. The winner is wrapped as a `CreativeBrief`: a list of `must_events`, a list of `must_not_events` ("Macbeth kills Duncan", "Banquo dies"), per-trait envelopes (Macbeth's `guilt` envelope is now $[0.1, 0.3]$ rather than $[0.6, 0.8]$), and a syuzhet window. The brief is the only object handed to the renderer.

C.8 Step 7: constrained rendering

The renderer LLM is invoked with the brief as system prompt plus a small style context (the synopsis's voice settings: synoptic-narration, third-person past tense, sparse density). It writes the banquet as a tense scene of overheard whispers, of Banquo unsuspecting and gracious, of Macbeth conspicuously unable to enjoy his elevation; the daggers stay sheathed; no ghost appears, because Banquo is alive. The renderer is free to choose dialogue, description, and pacing, but cannot invent a causal edge, shift a trait outside its envelope, or open a channel that does not exist. The output is a prose passage, not a graph.

C.9 Step 8: audit, re-extraction, and merge

The auditor runs a single categorical pass whose prompt bundles three check families. The *causal* family reverse-engineers the prose into (source, target, modality) triples and matches each against the brief; an LLM-emitted "Banquo's ghost rises behind Macbeth" would be flagged as a *miracle step* (no licensing edge), but in this run nothing of the kind appears. The *abduction* family runs counterfactual probes against the prose — "if Banquo had suspected the prophecy, would the prose still hold?" — and checks the answers against the brief's belief envelope. The *affective* family re-runs the four scorers on the prose and compares them to the brief's targets. If any pass fails the `engine_thresholds` gate or the `llm_pass` gate, the renderer is re-invoked with the structured feedback appended, up to `max_correction_retries` (default 3). On convergence, `ingestion` is run again on the prose alone, and the resulting graph is diffed against the brief; the diff becomes a new `VersionRow` with `world_id="shadow"` and `ancestor_id` pointing at the original counterfactual root. The user sees the new version appear in the sidebar in violet, can diff it against the factual mainline, and can `promote_branch` it to canon if desired.

C.10 What this walkthrough demonstrates

Three properties matter. **Inspectability:** every intermediate object — the ego-graph, the `CausalPhysicsResult`, the `CreativeBrief`, and the per-iteration `AuditCycleSnapshot.audit_result` (an `AuditResult`) inside the final `FeedbackLoopResult` — is a typed Pydantic value, persisted and visible in the UI. **Locality of LLM authority:** the LLM authors the initial ontology (Step 1), the candidate prose (Step 7), and the audit critique (Step 8); everything else is symbolic computation over typed code. **Branch safety:** the counterfactual is a sibling row, not a mutation; canon is preserved, the shadow is diffable and promotable, and the version DAG records the lineage.

D The authoring user interface

The authoring workspace is a NiceGUI single-page application (`shadow_loom_ui/app.py`) reachable on `http://localhost:7860`. It is organised as a left-hand *version sidebar*, a top-level *chat / command bar*, a dedicated *Answer panel* for read-only Q&A results, and nine cross-linked tabs that share a central `AppState` pub/sub event bus (`shadow_loom_ui/state.py`). The same active-version pointer is mirrored to the database so an MCP client and the UI always see the same tip. Every panel header carries a clickable info-icon help popover (`components/help_popover.py`) opening a structured Markdown reference for the surface — what the panel does, how to read its diagrams, what the controls mean, and what does *not* appear there — so the workspace doubles as its own documentation. This appendix walks through every surface in turn, with a concrete example session drawn from the bundled plot fixtures.

D.1 Version sidebar

The sidebar (`shadow_loom_ui/components/version_sidebar.py`) renders the project's full `VersionRow` DAG as a tree rooted at the original ingestion. Factual rows are green, shadow rows (counterfactual forks under `branch_policy=auto`) are violet. Toolbar actions: *Delete* re-parents children through `db.delete_version`; *Reparent* changes a row's `ancestor_id` (rejected if it would multi-root the tree); *Promote to canon* (shadow rows only) calls `db.promote_branch` to copy a shadow onto a fresh factual row; *Diff against factual* (shadow rows only) opens a structured diff against the current factual head. Clicking any row swaps the active version and resets every tab.

Example. After running the *Macbeth refuses the murder* counterfactual of App. C, the sidebar shows a green chain `ingestion` → `manual_edit` → `pipeline_run` for the factual mainline, with a single violet child branching off the `pipeline_run` row. Right-clicking the violet row offers *Diff* *against factual* (showing the absent EVT_DUNCAN_MURDER and the suppressed downstream deltas) and *Promote to canon* (which would replace the mainline).

D.2 Story tab

The STORY tab (components/story_tab.py) is the entry point: a plain-prose textarea plus a *Save & Re-ingest* button. Saving runs the full pipeline.run_pipeline and persists the result as a new version with source="ingestion". This tab is intentionally not autosaved: the textarea is prose, not graph, and clicking re-ingest is a deliberate action.

Example. Pasting the public-domain sample_plots/death_on_the_nile.txt synopsis and clicking *Save & Re-ingest* produces, after the ingestion fan-out described in App. C.2, a WorldStateV1 containing ENT_POIROT, ENT_LINNET, ENT_JACQUELINE, ENT_SIMON, the LOC_KARNAK steamer, and the conspiracy event EVT_PLAN_HATCHED with syuzhet_index placed late but fabula_time placed early — the very gap that the dramatic-irony scorer reads off in App. B.5.

D.3 Explorer tab

The EXPLORER tab (components/explorer_tab.py) is a read-only tree view of every node in the active world model, grouped by category, with an inspector panel for the selected node. It is the canonical way to confirm that ingestion has produced the right schema before running queries.

Example. Loading the Macbeth fixture and clicking ENT_MACBETH opens an inspector showing each TraitVector (ambition: 0.7, inertia 0.55, evidence strong; guilt: 0.1, inertia 0.25, evidence weak), each Belief with provenance, the state_timeline entries (one per major shock event), and the entity's outgoing causal and relationship edges. The inspector is the principal debugging tool when an extraction agent has produced an unexpected trait or belief.

D.4 World tab

The WORLD tab (components/world_tab.py) is a Cytoscape graph visualisation of entities, locations, objects, and edges, with a fabula-time slider beneath. Sliding the cursor through fabula time updates node colours and sizes via state.set_fabula_cursor (debounced 120 ms) so that the viewer can watch the social topology evolve.

Example. On Reservoir Dogs, scrubbing the slider from fabula $t = 0$ (the diner conversation) through to $t \approx 8000$ (the warehouse stand-off) animates the ENT_ORANGE node turning from neutral grey to red as the wound state advances; the spatial graph contracts as characters converge on the warehouse; and the trust edges from ENT_WHITE to ENT_ORANGE thicken. The same scrub on Romeo and Juliet shows Romeo's despair edge widening sharply over the final scene.

D.5 Causality tab

The CAUSALITY tab (components/causality_tab.py) has three sub-tabs. *Topology* renders a Sankey diagram of causal flow up to the fabula cursor: thicker ribbons for higher-weight edges, colour banded by causality_type. *Evolution* plots trait trajectories per entity over fabula time, with a vertical needle synced to the cursor. *Affective Dashboard* shows gauges and time series for the four structural scorers and the six emotion scorers, computed by viz_helpers.compute_affective_scores (which routes through DirectiveAssembler.compute_*_score). The top twenty entities by event degree are shown by default to keep the plots readable.

Example: Macbeth Evolution sub-tab. Selecting ENT_MACBETH, ENT_LADY_MACBETH, and ENT_MACDUFF and scrubbing through fabula time shows Macbeth's guilt curve climbing in two steps (after EVT_DUNCAN_MURDER and again after EVT_BANQUO_GHOST); Lady Macbeth's ruthlessness curve peaks early and then collapses into the sleepwalking scene; and Macduff's grief curve has a single step at the slaughter event. Switching to the *Affective Dashboard*, the suspense gauge sits high through Acts II–IV and zeroes out cleanly when the duel begins (the suspense–despair boundary built into the balance × stakes combiner: when $w_{\text{hope}} = 0$ nothing is left to author against the threat, and the score returns 0).

Example: Death on the Nile Affective Dashboard. The dramatic-irony gauge sits at ~ 0.7 from the moment the conspiracy is revealed to the audience and drops to zero only when Poirot delivers the denouement; the mystery gauge follows the inverse trajectory, peaking on the morning of the murder.

D.6 Reasoning tab

The REASONING tab (`components/reasoning_tab.py`) is where queries are typed and where intervention / counterfactual traces are inspected. The chat box accepts natural-language queries that are parsed into one of the typed query objects (`ObservationQuery`, `InterventionQuery`, `CounterfactualQuery`, `DirectiveQuery`, `InterrogationQuery`, `ManualEditQuery`, `EvaluationQuery`; a `GeneralQuery` fallback type also exists in the codebase but is no longer surfaced as an explicit chat mode — free-form questions are routed through `InterrogationQuery` instead). Write-mode queries route their result into this tab, where `reasoning_helpers.py` formats the `CausalPhysicsResult`: each mutation is shown with its (I, ι, Δ) triple, each blocked propagation with the reason (inertia, affordance, spatial), each abduction back-fill with the evidence weight, and any hidden deltas with the syuzhet window in which they will be revealed. Read-mode queries (`InterrogationQuery`) skip this tab entirely: they are dispatched through `shadow_loom/answer.py` and surface in the dedicated *Answer panel* above the chat bar (`components/answer_panel.py`) as a card with the model's claim, a confidence badge, an evidence-id list back to the graph nodes consulted, and any caveats. The Answer panel clears on `VERSION_CHANGED` and `PROJECT_LOADED` so a stale answer never lingers across versions; no `VersionRow` is written for a read-mode query.

Example. On Macbeth, typing *"If Macbeth had told Banquo about the prophecy on the heath, would Banquo have trusted him at the banquet?"* parses to a `CounterfactualQuery` with focal entities {ENT_MACBETH, ENT_BANQUO}. The trace panel shows the abduction step (Banquo's `trust` prior is inferred from the battlefield events), the do-step (a new `utterance` event is inserted on the heath channel), and the propagation step (Banquo's `suspicion` trait is suppressed at the banquet, but his `prudence` trait *raises* a new caution gate, which the engine reports as a non-trivial counterfactual outcome — not all secrets shared are secrets defused).

D.7 Audit tab

The AUDIT tab (`components/audit_tab.py`) displays the structured `AuditResult` for the most recently rendered scene. The three sections (causal, abductive, affective) each show their loss values and any offending prose spans, highlighted inline. A green badge indicates a passed pass; an amber badge indicates a warning (the prose is acceptable but drifted); a red badge indicates a hard failure that triggered a re-render.

Example. On Gone Girl, after generating a continuation of Amy's diary entry, the causal audit flags a hallucinated "Nick suddenly suspects Amy is alive" span with no licensing causal edge in the brief. The auditor's structured feedback is displayed alongside the offending span, and the user can either accept the auto-rerun or open the brief in the REASONING tab to widen the envelope.

D.8 Research tab

The RESEARCH tab (`components/research_tab.py`) is the optional open-web companion: the user manages a list of research topics, launches background lookup tasks (currently against Tavily; see §A.2), and browses the resulting facts, each carrying a source URL and a confidence score. The crucial design point is *segregation*: research is a separate store. Facts are not written into the world model automatically; they are reference material available to the author and to the renderer's prompt scaffolding (as a `BACKGROUND CONTEXT -- NOT AUTHORITATIVE` block) only. Promoting a research finding into canon is a deliberate, manual edit in the EDITOR tab. A status strip at the top of the panel shows in-flight lookups, the stored fact counts per topic, and the last refresh time.

Example. Authoring an alternate-history fork on the *Tinker, Tailor, Soldier, Spy* fixture, the user adds the topic *"Cambridge Five chronology, declassified MI6 sources"*, runs lookup, and reviews the returned facts before citing one as inspiration for a new `EventNode` in the EDITOR tab. The world model itself is unchanged until the user explicitly authors the new event; the research entry remains visible as provenance for the editorial decision.

D.9 Editor tab

The EDITOR tab (`components/editor_tab.py`) is the manual world-model editor. It has three integrated surfaces backed by a single JSON textarea (the source of truth): a *Validate* button that runs the same `_programmatic_validation` as ingestion; a *Structural editor* with one collapsible panel per collection (locations, entities, objects, events, world traits, causal edges, relationship edges, spatial edges, channels); and the raw JSON textarea itself for fields the structural editor does not expose (sparse trait values, beliefs, ambient state, evidence strengths). The save flow runs Pydantic validation, then `_programmatic_validation`; errors block the save with a dialog listing every issue, warnings present a *Save Anyway* confirmation, and on confirm the world is persisted as a new `VersionRow` with `source="manual_edit"` parented to the current row.

Example. Authoring the `example_worlds/macbeth.py` fixture from scratch, the user adds `LOC_HEATH` via the structural editor (typing *"the heath"* into the ID field auto-prefixes to `LOC_THE_HEATH`, edited to `LOC_HEATH`), then `ENT_MACBETH`, then the witches as a single multi-actor entity, then the prophecy event with `event_type= revelation` and `via_channel_id` pointing at a freshly added `Channel(participants=[ENT_MACBETH, ENT_BANQUO, ENT_WITCHES], medium="speech", intelligibility={...: 1.0})`. After each addition the validator panel updates: the green panel shows the running counts, an amber panel warns about the missing mutation coverage on `ENT_MACBETH.ambition` (no `state_timeline` yet), and a red panel reports the broken `addressee_ids` reference until the witches entity is added.

D.10 Export tab

The EXPORT tab (`components/export_tab.py`) is the egress surface: download the active world as a single `world_state.json`, download the version tree as a structured JSONL file, or copy the active version row ID for use with the MCP server or scripts.

Example. A researcher running cross-fixture comparisons exports each of the twenty bundled fixtures' `world_state.json` files, computes the entity-degree distribution and the per-fixture suspense time-series in a notebook, and persists the results without ever loading the UI's visualisation code.

D.11 Performance discipline

Three small disciplines keep the workspace responsive during long scrubs and rapid edits. (i) Cursor writes are debounced through `state.set_fabula_cursor`, which schedules an emit on a 120 ms tail. (ii) Heavy panels are gated by `is_path_visible(path)` so that off-tab rebuilds are skipped. (iii) `state.spawn_panel_task(panel_id, coro)` cancels any prior task with the same `panel_id`, collapsing rapid scrubs to the most recent render; ECharts panels build the chart once and patch options in place.

D.12 Cross-tab event bus

Six events flow through `AppState`: `PROJECT_LOADED` (emitted by the project picker and the MCP `open_project` tool, listened to by every tab); `WORLD_STATE_CHANGED` (emitted by `load_db_version`, manual save, and pipeline runs, triggering cache invalidation in `viz_helpers`); `VERSION_CHANGED` (sidebar highlight + raw-text reload); `FABULA_CURSOR_CHANGED` and

SYUZHET_CURSOR_CHANGED (cursor sliders); and ACTIVE_PATH_CHANGED (top tab change, gating heavy panels). Adding a new tab is therefore a matter of subscribing to the relevant subset of these events and rendering against the current AppState.

D.13 Why a UI matters for a research artefact

The UI is not an optional consumer surface; it is the principal debugging tool for the symbolic layers. The version sidebar makes branch lineage visible; the Causality tab makes the affective scorers' trajectories visible; the Reasoning tab makes the causal-physics trace visible; and the Audit tab makes the brief-vs-prose mismatch visible. A reviewer or collaborator can load any of the twenty bundled fixtures, run the counterfactuals discussed in App. B, and inspect every intermediate object without writing a line of code. That inspectability is what the typed-substrate design was meant to buy; the UI is where it is realised.